ANATOMY · ORGANS OF THE ABDOMINAL CAVITY

Liver, Biliary Tree, Pancreas & Spleen

High-Yield Study Guide

Projection · Development · Lobes & segments · Vessels & lymph · Portal hypertension · Bile ducts · CALOT's triangle · Pancreatic ducts · Splenic circulation

Source: ClinicalKey reading "Organs of the abdominal cavity", sections 6.5 (Liver and gallbladder) and 6.6 (Pancreas, Spleen), Figs 6.88–6.143. Section and figure numbers in [brackets] beside each heading let you find the matching page in the reading.

Contents

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How to read this guide

→	"leads to / drains to / gives rise to" — follow the arrow to connect ideas
[§ / Fig]	Where the topic appears in the reading
Gold box	Key numbers and must-know facts from the reading
Green box	Memory aid / mnemonic
Blue box	Clinical remarks from the reading (and clinical notes marked "not from reading")
Red box	Source note — a statement in the reading that needs care, a contradiction or a typo
(not from reading)	Added for context or memory; not stated in the reading

Study guide built from the reading only; added material is labelled "not from reading". Figures are reproduced from the reading. The reading uses Latin anatomical terms; this guide gives the Latin term with the English equivalent the first time it appears.

PART 2 Glossary (Latin ↔ English)

Latin term	English / meaning
Hepar · Vesica biliaris	Liver · gallbladder
Splen [Lien] · Pancreas	Spleen · pancreas
Facies diaphragmatica / visceralis	Diaphragmatic (upper, convex) / visceral (lower, facing the viscera) surface
Margo inferior	Inferior (anterior lower) border of the liver
Area nuda	Bare area of the liver → fused with the diaphragm, no peritoneum
Lobus dexter / sinister	Right / left anatomical lobe
Lobus quadratus / caudatus	Quadrate lobe (ventral) / caudate lobe (dorsal), both on the visceral surface
Lig. falciforme hepatis	Falciform ligament → separates the anatomical lobes in front
Lig. coronarium · Lig. triangulare	Coronary ligament · right and left triangular ligaments → attach the liver to the diaphragm
Appendix fibrosa hepatis	Fibrous appendix → pointed tip of the left triangular ligament
Lig. teres hepatis	Round ligament → remnant of the fetal umbilical vein (V. umbilicalis)
Lig. venosum (ARANTII)	Ligamentum venosum → remnant of the fetal ductus venosus
Porta hepatis	Hilum of the liver → portal vein, proper hepatic artery, common hepatic duct
Omentum minus	Lesser omentum → liver to stomach and duodenum (from the dorsal mesohepaticum)
Lig. hepatoduodenale	Hepatoduodenal ligament → carries portal vein, hepatic artery, common bile duct
GLISSON's triad	Portal triad → branch of portal vein + hepatic artery + bile duct in connective tissue
V. centralis · Vv. sublobulares	Central vein of the lobule · sublobular veins → hepatic veins
Vv. hepaticae	Hepatic veins (right, intermediate/middle, left) → inferior vena cava
V. portae hepatis	Hepatic portal vein
A. hepatica communis / propria	Common / proper hepatic artery
Truncus coeliacus	Coeliac trunk
A. / V. mesenterica superior / inferior	Superior / inferior mesenteric artery / vein
A. / V. splenica	Splenic artery / vein
A. gastroduodenalis	Gastroduodenal artery
A. cystica · V. cystica	Cystic artery · cystic vein (gallbladder)
Vv. paraumbilicales	Paraumbilical veins → run in the round ligament to the umbilicus
Caput medusae	Dilated superficial periumbilical veins in portal hypertension
Corpus / Fundus / Collum vesicae biliaris	Body / fundus / neck of the gallbladder
Ductus cysticus · Plica spiralis (HEISTERI)	Cystic duct · spiral valve of Heister
Ductus hepaticus communis	Common hepatic duct (right + left hepatic ducts)
Ductus choledochus	Common bile duct (common hepatic + cystic duct)
Ampulla hepatopancreatica	Hepatopancreatic ampulla (of Vater) → shared end of bile and pancreatic ducts
Papilla duodeni major (VATERI) / minor	Major duodenal papilla (Vater) / minor papilla

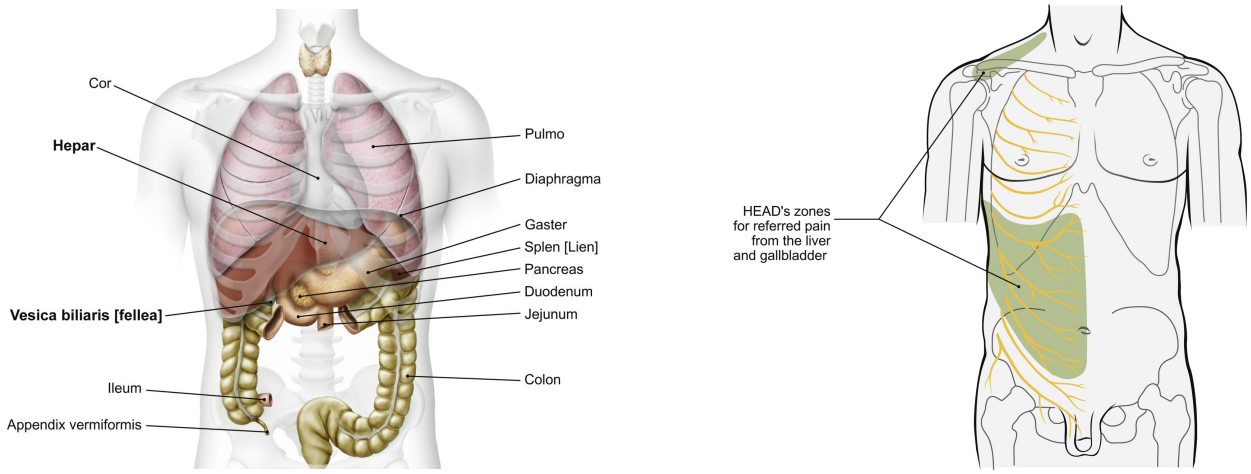
Latin term	English / meaning
M. sphincter ampullae (ODDI)	Sphincter of Oddi
Trigonum cholecystohepaticum	CALOT’s (hepatobiliary) triangle
Caput / Collum / Corpus / Cauda pancreatis	Head / neck / body / tail of the pancreas
Proc. uncinatus	Uncinate process → hooks behind the superior mesenteric vessels
Ductus pancreaticus (WIRSUNG)	Main pancreatic duct
Ductus pancreaticus accessorius (SANTORINI)	Accessory pancreatic duct → minor papilla
Insulae pancreaticae (LANGERHANS)	Islets of Langerhans → endocrine pancreas
Pancreas anulare / divisum	Annular pancreas / pancreas divisum
Bursa omentalis · Recessus splenicus	Omental bursa (lesser sac) · its splenic recess
Lig. gastrosplenicum / splenorenale	Gastrosplenic / splenorenal ligament
Lig. phrenicocolicum	Phrenicocolic ligament → floor of the splenic niche
Hilum splenicum · Pulpa splenica	Splenic hilum · splenic pulp (red + white)
PALS	Periarterial lymphatic sheath → T cell zone of the white pulp
Nodi lymphoidei …	Lymph nodes (hepatici, coeliaci, phrenici, splenici, pancreatici …)
HEAD’s zone	Skin area (dermatome) where pain from an organ is referred
N. phrenicus · Plexus coeliacus	Phrenic nerve (C3–C5) · coeliac plexus
Truncus vagalis anterior / posterior	Anterior / posterior vagal trunk

PART 3 Content by Topic

A. LIVER

3.1 Projection and Referred Pain [§6.5.1 · Figs 6.88–6.89]

- Liver and gallbladder → **intraperitoneal**, right upper abdomen
- Upper margin of the liver → **right 4th intercostal space**; on the left slightly lower → **5th rib**
- Gallbladder fundus → **right midclavicular line at the 9th rib**
- Left lobe → left epigastrium (roughly to the left midclavicular line), **anterior to the stomach**
- Position moves with breathing → **down on inspiration, up on expiration** (attached to the diaphragm); also depends on lung size
- Domed diaphragm → front and back of the liver are partly **covered by the pleural cavity**
- Lower margin → follows the **costal margin** in the midclavicular line → a normal liver is **not palpable**



Left: projection of the abdominal organs onto the body surface (Fig 6.88). Right: HEAD's zones of the liver and gallbladder — T8–T11 on the right plus C4 over the right shoulder (Fig 6.89)

HEAD's zones (referred pain)

- Liver and gallbladder → dermatomes **T8–T11, right side**
- Also **C4 over the right shoulder** → phrenic nerve (mainly C4) → its **R. phrenicoabdominalis** supplies the peritoneum on the liver and gallbladder

CLINICAL REMARKS (reading)

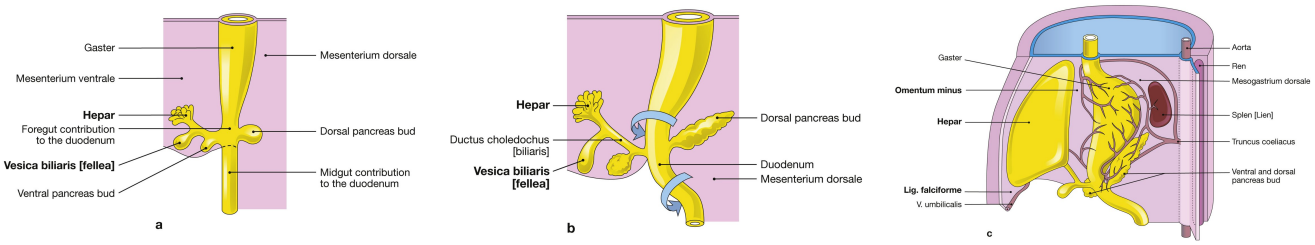
- Hepatitis, cholecystitis or a liver biopsy → **referred pain in the right shoulder**
- Liver size and consistency are part of every physical exam → early clue to steatosis (diabetes, alcohol), hepatitis or cirrhosis
- Inferior margin → **palpation during inspiration**; superior margin → **percussion** of the chest
- Rule of thumb → liver ≤ **12 cm** craniocaudal in the right midclavicular line

MEMORY AID (not from reading)

- Liver top = "**4 right, 5 left**" · Gallbladder = "**9 o'clock in the MCL**" (9th rib, midclavicular line)
- Shoulder pain from the liver → "**C3, 4, 5 keeps the diaphragm alive**" — and the phrenic nerve also feels the liver capsule

3.2 Development of Liver and Gallbladder [§6.5.2 · Fig 6.90]

- Epithelium of liver and gallbladder → **endoderm of the foregut** at the level of the future duodenum
- **Week 4 (from day 22)** → endoderm thickens into the **hepatic diverticulum**
  - Superior part → **liver primordium**
  - Inferior part → **bile duct system** (and gallbladder)
- Liver epithelium grows into the **septum transversum** → supplies the connective tissue, the sinusoids, and islands of **haematopoiesis**
- Liver moves into the **ventral mesogastrium** → splits it into two parts:
  - **Ventral mesohepaticum** → **falciform ligament** (to the anterior trunk wall)
  - **Dorsal mesohepaticum** → **lesser omentum** (liver to stomach and duodenum)



Development of the liver and gallbladder in weeks 4–5: (a) hepatic diverticulum and pancreatic buds, (b) during gut rotation, (c) liver in the ventral mesogastrium. Fig 6.90

3.3 Liver Overview — Surfaces, Lobes, Ligaments [§6.5.3 · Figs 6.91–6.92]

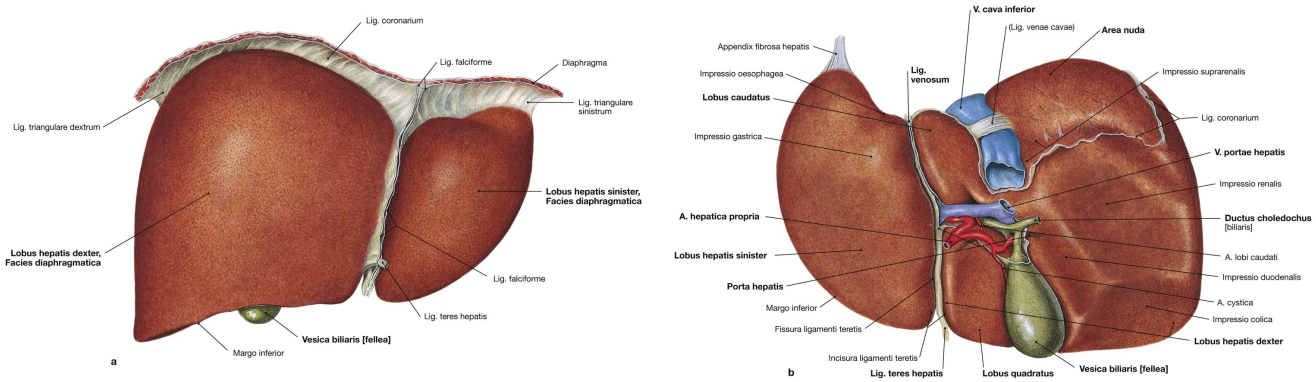
- **Largest gland** of the body (**1,200–1,800 g**) and the main metabolic organ
- **Diaphragmatic surface** → against the diaphragm; partly fused with it → **bare area (Area nuda)**, no visceral peritoneum
- **Visceral surface** + sharp **inferior margin** → face the abdominal viscera
- Anatomical lobes → large **right** and small **left**, separated in front by the **falciform ligament**

Ligaments

- Falciform ligament → continues upward as the **coronary ligament** → ends right and left as the **triangular ligaments** (to the diaphragm)
- Left triangular ligament → **Appendix fibrosa hepatis**
- **Round ligament (Lig. teres hepatis)** → remnant of the fetal **umbilical vein** → joins the lower edge of the falciform ligament; both reach the anterior trunk wall
- **Ligamentum venosum (ARANTII)** → remnant of the fetal **ductus venosus**

Visceral surface — the “H” around the porta hepatis

- **Porta hepatis** → in: **portal vein** + **proper hepatic artery**; out: **common hepatic duct**
- Left limb → fissure for the round ligament (front) and ligamentum venosum (back)
- Right limb → **IVC groove** (above/back) and **gallbladder fossa** (below/front)
- Between them → **quadrate lobe** (ventral, next to the gallbladder) and **caudate lobe** (dorsal, next to the IVC)



Left: liver, ventral view — falciform ligament, round ligament, right and left lobes (Fig 6.91a). Right: visceral surface — porta hepatis, quadrate and caudate lobes, gallbladder, IVC (Fig 6.91b)

KEY FACTS

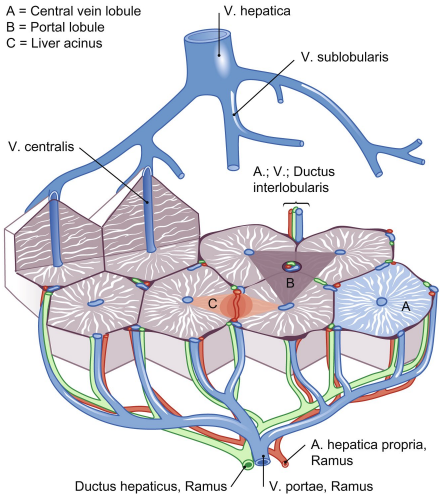
- **Four areas not covered by peritoneum** → bare area · porta hepatis · gallbladder bed · IVC groove
- Impressions of neighbouring organs on a fixed liver → **fixation artefacts**; in life the liver is soft and moulds to its neighbours (useful only to show position)

MEMORY AID (not from reading)

- Fetal remnants: **teres = umbilical vein** (“**T**he cord”), **venosum = ductus venosus** (same word)
- Visceral surface looks like an **“H”**: left bar = teres + venosum; right bar = gallbladder + IVC; crossbar = porta hepatis; **Q**uadrate in front, **C**audate behind (“**Q**uick in front, **C**aboose at the back”)

3.4 Microscopic Structure — Lobule, Portal Lobule, Acinus [§6.5.4 · Figs 6.93–6.94]

- Vessels and duct enter at the porta hepatis → branch in connective tissue → **GLISSON’s triad** in the periportal field
- **Hepatic veins** → run **separately** from the triad → drain to the IVC
- **Classic hepatic lobule** → almost **hexagonal**, radial plates (trabeculae) of hepatocytes, **central vein** in the middle, **portal tracts at 3–6 corners**
- Portal tract (GLISSON’s triad) → **interlobular artery + interlobular vein + interlobular bile duct** in connective tissue
- Blood flow → peripheral (portal) arteries and veins → **sinusoids** between hepatocyte plates → **central vein** → **sublobular veins** → **hepatic veins** → IVC
- Hepatocytes → take up nutrients, remove waste, secrete products (e.g. **plasma proteins**) into the blood
- Bile → flows the **opposite way**, between hepatocytes toward the portal fields → interlobular bile duct → common hepatic duct



Hepatic parenchyma: (A) classic lobule around the central vein, (B) triangular portal lobule around a portal tract, (C) rhomboid liver acinus. Fig 6.94

Unit	Shape	Centre	Corners / borders	Emphasises
Classic lobule	Hexagonal	Central vein	Portal tracts at 3–6 corners	Blood drainage
Portal lobule	Triangular	Bile duct (portal tract)	3 central veins	Bile secretion
Liver acinus	Rhomboid	Line joining 2 portal fields	2 portal fields + 2 central veins	O_2/nutrient supply → zones

KEY FACTS

- In the acinus, oxygen and nutrients are **best along the line between adjacent portal fields** → hepatocytes in different zones have **different jobs**

CLINICAL REMARKS (reading)

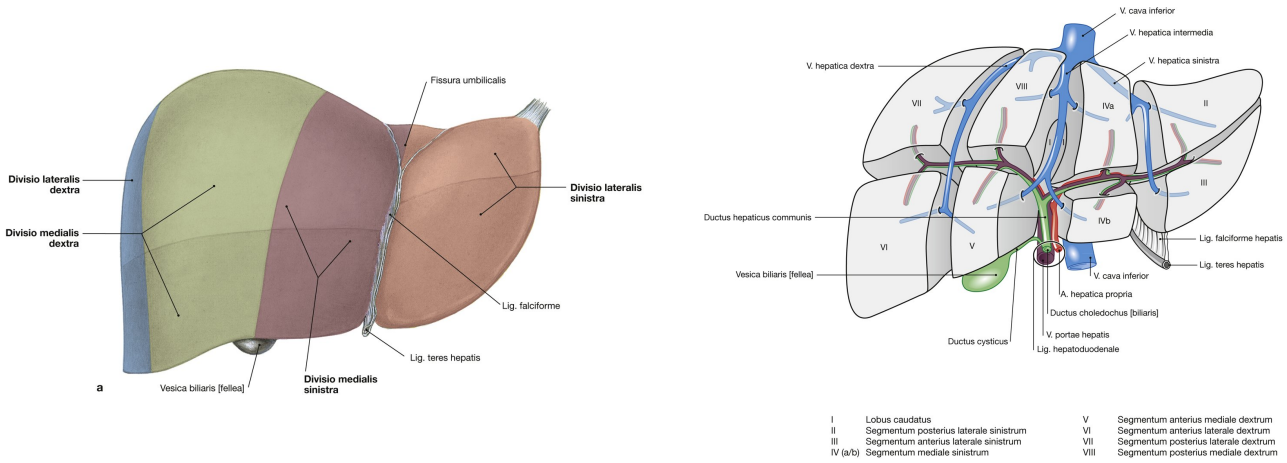
- **Cirrhosis** → nodular connective tissue remodelling destroys the lobules → blood flow impaired
- High parenchymal resistance → **portal hypertension** → **portocaval anastomoses** open (see 3.9)

CLINICAL CORRELATION (not from reading)

- Acinar zone 1 (periportal) is best oxygenated; zone 3 (around the central vein) is worst → zone 3 is hit first by ischaemia and by toxins such as paracetamol overdose

3.5 Liver Segments (Couinaud) [§6.5.5 · Figs 6.95–6.97]

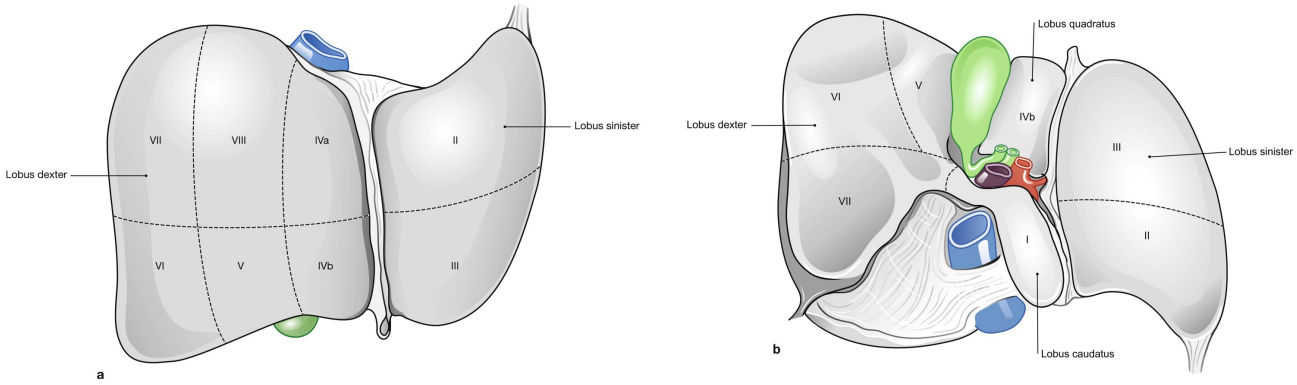
- **Three near-vertical hepatic veins** (right, intermediate, left) → divide the liver into **four divisions**:
  - **Left lateral division** → = anatomical left lobe, up to the falciform ligament, behind the left hepatic vein
  - **Left medial division** → between falciform ligament and gallbladder, at the level of the intermediate hepatic vein
  - **Right medial** and **right lateral divisions** → separated by the right hepatic vein; **no surface landmark**
- Branches of the **portal triad** → organise these into **8 functional segments**, each with its own portal vein + artery + bile duct branch → **functionally independent**
- Two segments at a time → loosely combined into the four divisions
- An old **“segment IX”** for part of the caudate lobe → **abandoned**



Left: the four divisions and eight segments shown in colour, ventral view (Fig 6.95a). Right: schematic of segments with the hepatic veins and portal triad branches (Fig 6.96)

Functional right vs left liver

- Segments **II–IV** → left branches of the portal triad → **functional left lobe**
- Segments **V–VIII** → right branches → **functional right lobe**
- Border between them → sagittal **IVC–gallbladder plane** (CANTLIE's line, not named in the reading), **not** the falciform ligament
- **Segment I = caudate lobe** → supplied by branches from **both** sides; drains **directly into the IVC** via short hepatic veins (§6.5.8)



Segments I–VIII marked with Roman numerals on the liver surface: (a) ventral, (b) dorsocaudal view. Fig 6.97

Segment	Location	Functional lobe
I	Caudate lobe	Both sides (independent)
II	Left lateral, superior (posterior)	Left
III	Left lateral, inferior (anterior)	Left
IV (IVa/IVb)	Left medial (includes quadrate lobe)	Left
V	Right anterior, inferior	Right
VI	Right posterior, inferior	Right
VII	Right posterior, superior	Right
VIII	Right anterior, superior	Right

Segment positions follow the standard Couinaud scheme shown in Figs 6.96–6.97 (positions not spelled out in the text).

CLINICAL REMARKS (reading)

- Segments matter in **visceral surgery** → resection along segment borders gives **little blood loss**
- Liver metastases → can be removed segment by segment in different parts of the liver without losing overall liver function
- Surgeon finds a segment by **ligating its vessel branch** → the segment **changes colour** (no perfusion)

HEADS-UP — SOURCE NOTE

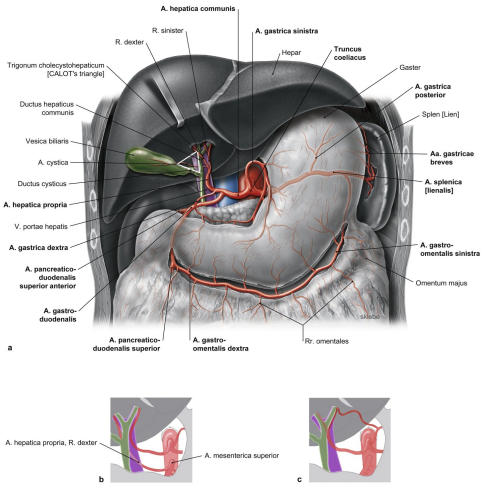
- The reading is inconsistent about segment I: Fig 6.96 says the caudate lobe is **not part of either** functional lobe; Fig 6.97 says it **functionally belongs to the left lobe**. Both agree it gets blood from **both sides**. Safest answer → **segment I is autonomous** (dual supply, own drainage to the IVC).

MEMORY AID (not from reading)

- Number the segments **clockwise** from the caudate lobe (I) when the liver is viewed from the front: II–III left lateral, IV left medial, then V–VIII round the right lobe (V lower front → VI lower back → VII upper back → VIII upper front)
- Functional border = "**Cava to Cyst**" (IVC to gallbladder), not the falciform

3.6 Arteries of the Liver and Gallbladder [§6.5.6 · Fig 6.98]

- Dual blood supply:
  - **Portal vein → 75 %** of liver blood flow → **nutrient-rich**
  - **Proper hepatic artery → 25 %** → **oxygen-rich**
- Arterial path → **coeliac trunk → common hepatic artery** → gives the **right gastric artery** → continues as the **proper hepatic artery**
- Proper hepatic artery runs in the **hepatoduodenal ligament** with the **portal vein** and **common bile duct** → to the porta hepatis
- At the porta → divides into a **right** and a **left branch**
- **Cystic artery** (to the gallbladder) → from the **right branch**



Arteries of the liver and gallbladder: (a) textbook pattern (~50 %), (b) contribution from the superior mesenteric artery to the right lobe, (c) contribution from the left gastric artery to the left lobe. Fig 6.98

KEY NUMBERS

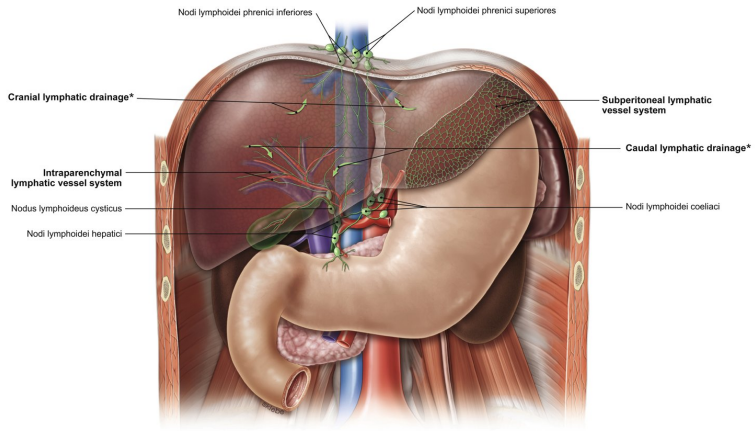
- Textbook pattern → only **~50 %** of people
- **SMA** helps supply the **right** lobe → **10–20 %** · **Left gastric artery** helps supply the **left** lobe → **10–20 %**
- Common hepatic artery arising from the SMA → rare, **1.5–4 %**

CLINICAL CORRELATION (not from reading)

- An aberrant right hepatic artery from the SMA runs behind the portal vein / bile duct → must be recognised in cholecystectomy and pancreatic head surgery

3.7 Lymph Vessels of the Liver and Gallbladder [§6.5.7 · Fig 6.99]

- Two lymph vessel systems:
  - **Subperitoneal** → on the liver surface
  - **Intraparenchymal** → follows the portal triad to the hilum
- Two **major** drainage routes:
  - **Caudal → hilum (80 % of lymph)** → hepatic nodes → coeliac nodes → **intestinal trunk**
  - **Cranial → through the diaphragm** → inferior and superior phrenic nodes → anterior and posterior mediastinal nodes → **bronchomediastinal trunks**
- Two **minor** routes:
  - To the anterior trunk wall → lymph vessels in the **round ligament** → **inguinal and axillary** nodes
  - From the left lobe → to the **stomach and pancreas**
- Gallbladder → its own **cystic node (Nodus lymphoideus cysticus)** at the neck → hilar nodes (mainly caudal)



Lymph vessels and nodes of the liver and bile ducts; arrows show cranial and caudal drainage. Fig 6.99

CLINICAL REMARKS (reading)

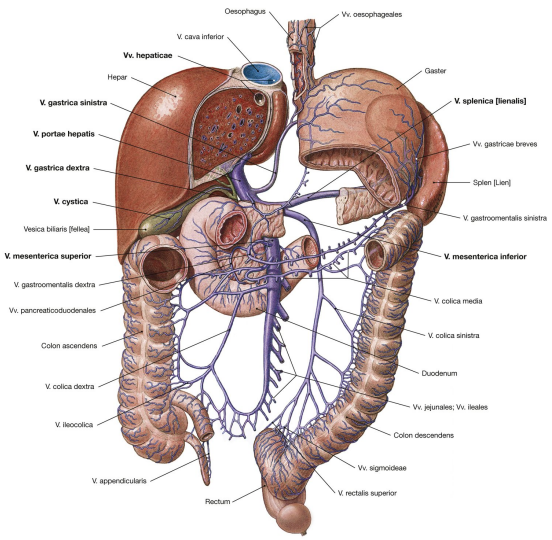
- Cranial route → liver carcinoma can metastasise to **thoracic lymph nodes**
- Rare spread to the anterior abdominal wall → painless, growing **umbilical nodule** = **Sister MARY JOSEPH node** → poor prognosis; diagnosis can be suspected on inspection
- Like **VIRCHOW's node** (left supraclavicular), you need to know lymph routes to trace the primary tumour

3.8 Veins of the Liver — Portal Vein and Hepatic Veins [§6.5.8 · Fig 6.100]

- Liver has an **incoming** and an **outgoing** venous system
- **Incoming → portal vein** → nutrient-rich blood from the **unpaired abdominal organs** (stomach, intestines, pancreas, spleen) → into the sinusoids together with arterial blood
- **Outgoing → three hepatic veins → IVC**; only the **caudate lobe** drains **directly** into the IVC through short hepatic veins

Formation of the portal vein

- **SMV + splenic vein** unite **behind the head (neck) of the pancreas** → portal vein
- **IMV** → into the **splenic vein (70 %)** or the **SMV (30 %)**
- Portal vein → about **8 cm** long → divides into **right (1–2 cm)** and **left (3–4 cm)** branches



Veins of the liver and gallbladder: portal vein and its tributaries, hepatic veins draining to the IVC. Fig 6.100

Vein	Drains	Tributaries
Splenic vein	Spleen, parts of stomach and pancreas	Short gastric veins · left gastro-omental vein · pancreatic veins (tail and body)
Superior mesenteric vein	Parts of stomach and pancreas, entire small intestine, ascending + transverse colon	Right gastro-omental vein (with pancreaticoduodenal veins) · pancreatic veins (neck, body) · jejunal + ileal veins · ileocolic · right colic · middle colic
Inferior mesenteric vein	Descending colon, upper rectum	Left colic · sigmoid veins · superior rectal vein (connects with middle + inferior rectal veins → IVC territory)
Direct to portal vein	—	Cystic vein · paraumbilical veins (via round ligament) · right and left gastric veins · posterior superior pancreaticoduodenal vein

MEMORY AID (not from reading)

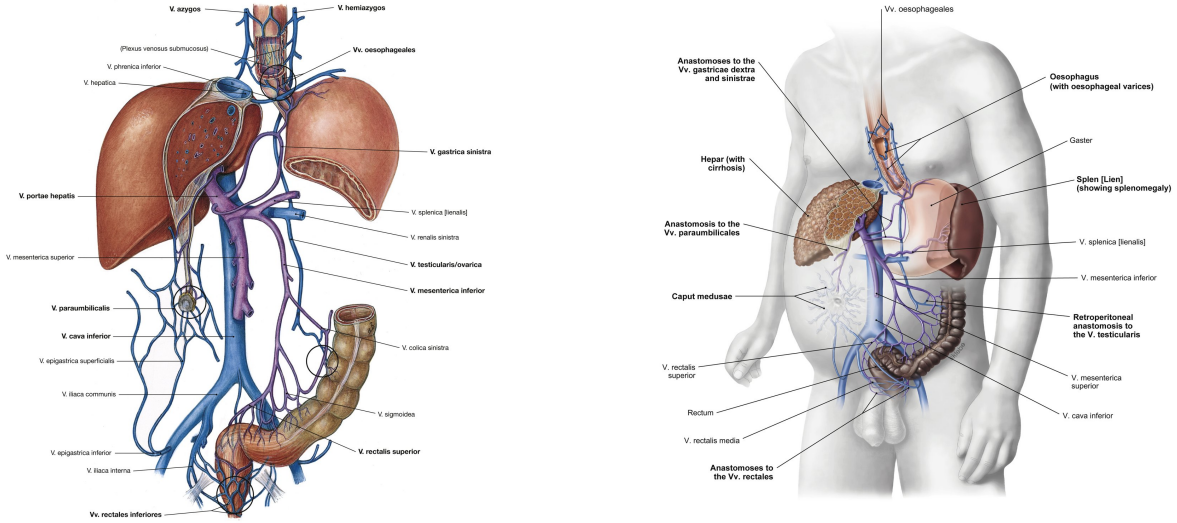
- Portal vein = "**S + S**" (**S**plenic + **S**MV) behind the pancreatic neck; IMV joins the splenic vein "**7 out of 10**" times
- Direct tributaries = the "**C-PGP**" veins → **C**ystic, **P**araumbilical, **G**astric (L+R), **P**ancreaticoduodenal (post. sup.)

3.9 Portocaval Anastomoses, Cirrhosis and Portal Hypertension [§6.5.9–6.5.11 · Figs 6.101–6.103]

- Portocaval anastomoses → connections between **portal vein** tributaries and **superior/inferior vena cava** tributaries → let portal blood **bypass the liver**

Site	Portal side	Caval side	Result when dilated	Clinical weight (reading)
1. Oesophagus	Right + left gastric veins	Oesophageal veins → azygos veins → SVC	**Oesophageal varices** (submucosal)	**Most important** → rupture = life-threatening haemorrhage; commonest cause of death in cirrhosis
2. Umbilicus	Paraumbilical veins (round ligament)	Deep: sup. + inf. epigastric veins · superficial: thoracoepigastric + superficial epigastric veins → SVC/IVC	**Caput medusae**	Diagnostic only; rare but unmistakable
3. Rectum	Superior rectal vein (IMV)	Middle + inferior rectal veins → internal iliac → IVC	Rectal venous dilation	Not clinically significant

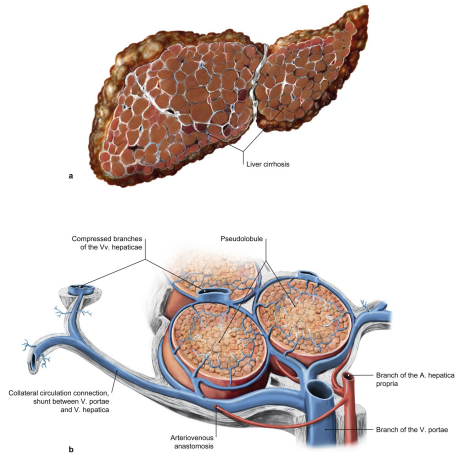
Site	Portal side	Caval side	Result when dilated	Clinical weight (reading)
4. Retroperitoneum	IMV (and other portal tributaries)	Testicular / ovarian veins → IVC	Retroperitoneal collaterals	Not clinically significant



Left: portocaval anastomoses (black circles); caval tributaries blue, portal tributaries purple (Fig 6.101). Right: signs of portocaval anastomoses in cirrhosis (Fig 6.103)

Liver cirrhosis

- **End stage** of many chronic liver diseases → liver "**scarred**" by slow inflammation or persistent damage (unlike acute destruction, e.g. **death-cap mushroom** poisoning)
- Causes → **worldwide: viral hepatitis B, C, D** · **industrialised countries: alcohol**, and increasingly **steatosis in diabetes or obesity**
- Macroscopic → **nodules** on the liver surface
- Microscopic → normal lobular architecture lost → **pseudonodules** + more connective tissue → **veins compressed**
- Chain → high parenchymal resistance → **portal venous stasis** → **portal hypertension** → pre-existing portocaval connections **open or recanalise**



Cirrhosis: (a) macroscopic nodules, frontal section; (b) pseudonodule formation with compressed vessels. Fig 6.102

KEY FACTS

- Ruptured **oesophageal varices** → **most common cause of death** in cirrhosis
- **Caput medusae** → rare, but "cirrhosis cannot be overlooked" when present
- Rectal and retroperitoneal anastomoses → **not clinically significant** (per the reading)

HEADS-UP — SOURCE NOTE

- The reading says “four” anastomoses but prints the rectal and retroperitoneal routes in one bullet. Count them as **four**: oesophageal, paraumbilical, rectal, retroperitoneal.

MEMORY AID (not from reading)

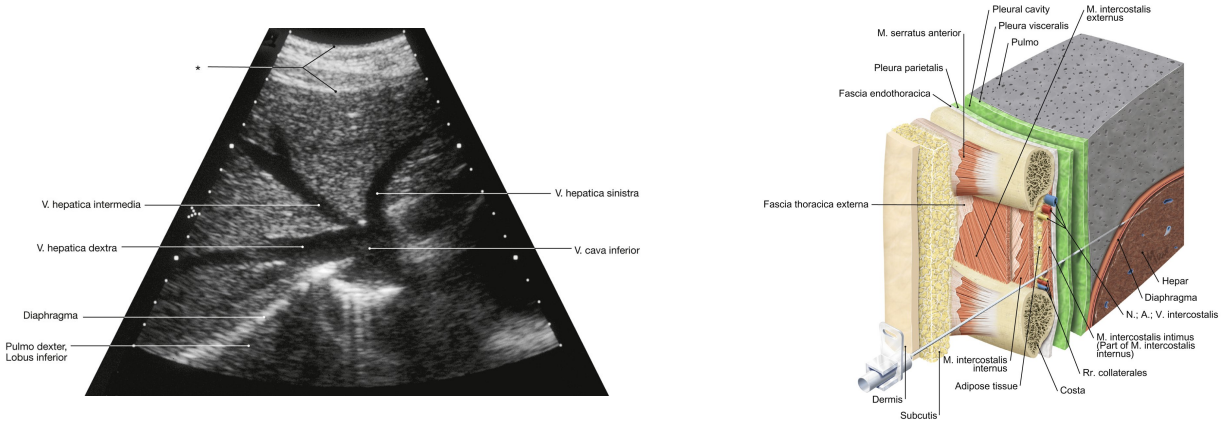
- Portal hypertension shows up at the “**top, middle, bottom and back**” → oesophagus, umbilicus, rectum, retroperitoneum
- Danger ranking: **oesophagus kills**, **umbilicus tells**, rectum and back don’t matter much

3.10 Liver Imaging and Biopsy [§6.5.12–6.5.13 · Figs 6.104–6.107]

- **Ultrasound** → standard, non-invasive look at the parenchyma → can show steatosis, fibrosis (hepatitis, cirrhosis), focal tumours and cysts
- Hepatic veins seen joining the IVC; portal vein branches seen in the parenchyma (Figs 6.104–6.105)
- Unclear ultrasound → **liver biopsy** or **laparoscopy**

Percutaneous liver biopsy

- Done under **ultrasound**, in **expiration**, through a **lower intercostal space**
  - Why expiration and low ICS → the liver is covered from above by the **pleural cavity** → risk of **pneumothorax**
- Needle passes at the **upper border of the rib below** (“superior costal margin”) → spares the intercostal vessels and nerve
- Liver capsule peritoneum → **phrenic nerve (C3–C5)** → patients often feel **right shoulder pain**
- Indications → nature of unclear tumours; **staging** of hepatitis or cirrhosis; definitive diagnosis needs **tissue** (pathologist)



Left: hepatic veins joining the IVC, ultrasound (Fig 6.104). Right: layers of the chest wall over the liver and needle path for liver biopsy (Fig 6.107)

HEADS-UP — SOURCE NOTE

- The reading writes “echo-dense steatosis (low echo) or fibrosis (increased echo)”. Standard teaching (not from reading): a **fatty liver is hyperechoic** (bright) on ultrasound; fibrosis gives a coarse, heterogeneous echo pattern. Treat this sentence with care.

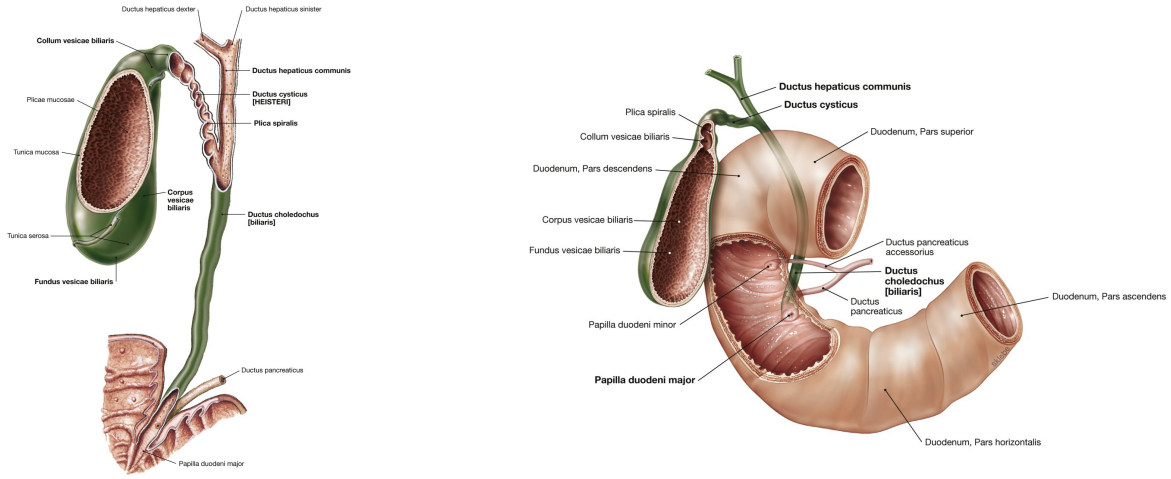
MEMORY AID (not from reading)

- Intercostal vessels and nerve run in the groove under each rib → “**VAN under the rib, needle over the rib**”

B. GALLBLADDER AND BILE DUCTS

3.11 Gallbladder and Extrahepatic Bile Ducts [§6.5.14–6.5.16 · Figs 6.108–6.112]

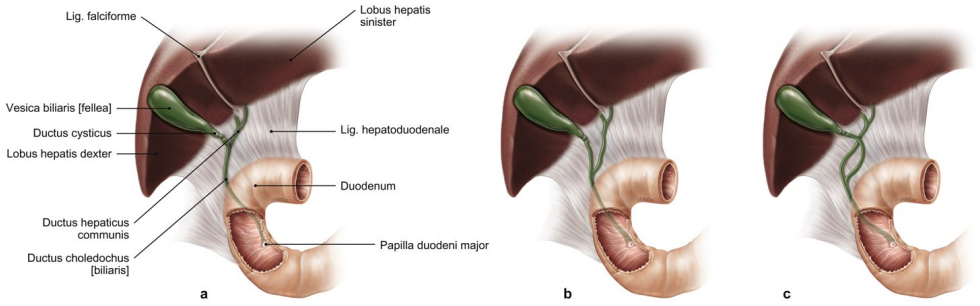
- Gallbladder → holds about **40–70 ml** of bile
- Parts → **fundus** · **body** · **neck** → **cystic duct**, closed by the **spiral fold (HEISTER)**
- Duct path:
  - Right + left hepatic ducts → **common hepatic duct** (formed in the liver)
  - Common hepatic duct + cystic duct (in the hepatoduodenal ligament) → **common bile duct (Ductus choledochus)**
- Common bile duct course → in the hepatoduodenal ligament **in front of the portal vein** → **behind the superior part of the duodenum** → through/behind the **head of the pancreas** → **descending duodenum**
- Usually joins the **pancreatic duct** → opens on the **major duodenal papilla (VATER)**



Left: gallbladder and extrahepatic bile ducts, ventral view (Fig 6.108). Right: gallbladder, bile ducts and duodenum (Fig 6.111)

KEY NUMBERS

- Gallbladder volume → **40–70 ml**
- Cystic duct → **~3 mm** wide, **2–4 cm** long
- Common bile duct → **~6 cm** long, **0.4–0.9 cm** wide · **> 1 cm** suggests **cholestasis**



Variants of the junction of the common hepatic duct and cystic duct: (a) high junction, (b) low junction, (c) low junction with crossover. Fig 6.112

CLINICAL REMARKS (reading)

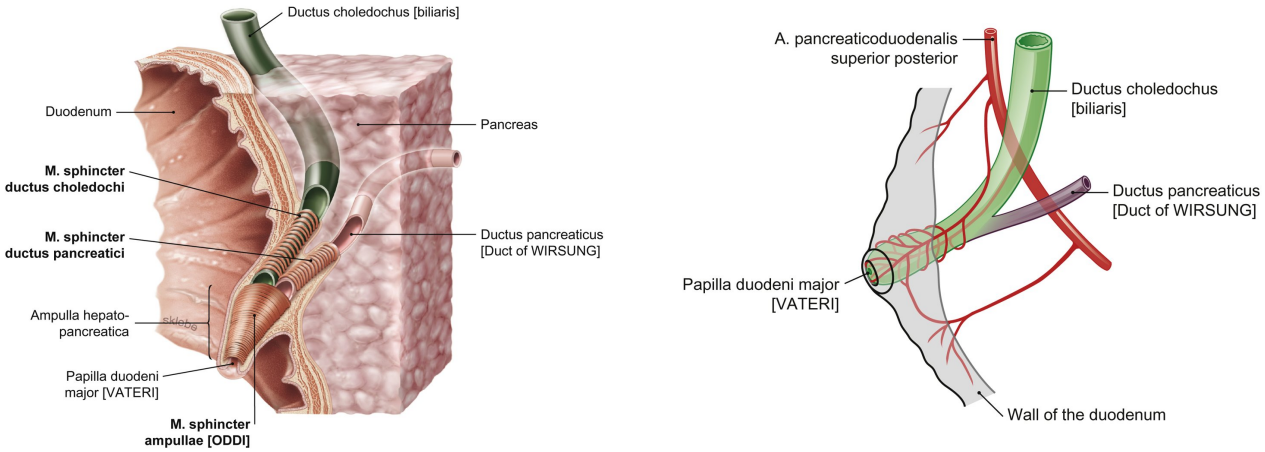
- **Cholecystitis** → serious; infection can spread by the blood → **sepsis**
- Treatment of choice → **cholecystectomy within 24 h** of diagnosis, done **laparoscopically**
- Duct variants must be known in **gallstones (cholecystolithiasis)** and cholecystectomy
- Duct imaging → **ERCP** (endoscopic retrograde cholangiopancreatography)
- **Laparoscopy** (Figs 6.109–6.110) → inspect the whole abdominal cavity, remove the gallbladder, or take **targeted liver biopsies** when imaging and blind biopsy fail

3.12 Ampulla Hepatopancreatica and Sphincters [§6.5.17 · Figs 6.113–6.114]

- In **60 %** → common bile duct + pancreatic duct join to form the **hepatopancreatic ampulla** → opens at the **major papilla (VATER)**
- Papilla position → **8–10 cm from the pylorus**, **dorsomedial wall** of the **middle third of the descending duodenum**
- Sphincter system under the mucosa:
  - **Sphincter of the common bile duct** (circular smooth muscle)
  - **Sphincter of the pancreatic duct**
  - **Sphincter of the ampulla (ODDI)** → distal parts around the ampulla

Blood supply of the common bile duct

- Upper parts → fine branches of the **cystic artery**, **right branch of the proper hepatic artery**, and ascending branches of the **gastroduodenal artery**
- **Distal third + ampulla → posterior superior pancreaticoduodenal artery**



Left: hepatopancreatic ampulla and its sphincters (Fig 6.113). Right: arterial supply of the ampulla and distal common bile duct (Fig 6.114)

CLINICAL REMARKS (reading)

- Gallstone leaves the gallbladder → **blocks the major papilla** → bile backs up (**cholestasis**) → usually a **painful, swollen gallbladder**
- Bilirubin deposits → yellow sclera and skin = **jaundice** → stone removed **endoscopically**; papilla is well perfused → risk of **strong bleeding**
- Cholestasis → always exclude **pancreatic head cancer** → here the enlarged gallbladder is **not inflamed → mostly painless**

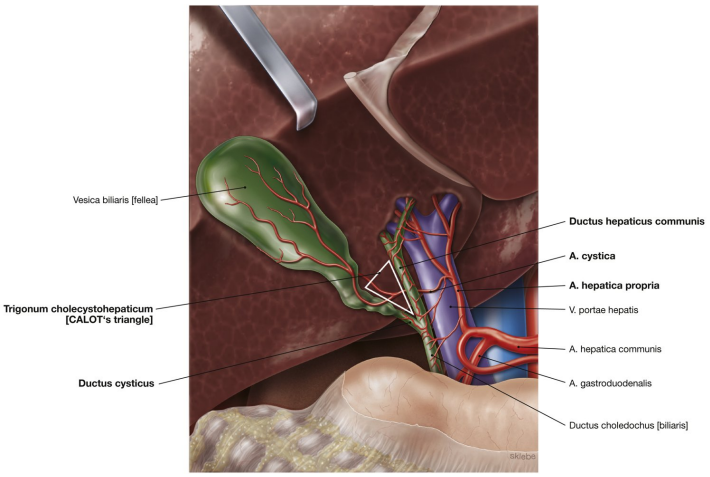
MEMORY AID (not from reading)

- Painless, palpable gallbladder + jaundice → think tumour, not stone (COURVOISIER's sign, not named in the reading)

3.13 CALOT’s Triangle and Biliary Imaging [§6.5.18–6.5.19 · Figs 6.115–6.117]

- **CALOT’s triangle (Trigonum cholecystohepaticum)** → formed by the **cystic duct**, **common hepatic duct** and **inferior surface of the liver**
- In **75 %** → the **cystic artery** arises inside this triangle from the **right branch of the proper hepatic artery** → runs behind to the cystic duct and gallbladder neck

Direction	Border of CALOT’s triangle
Cranial	Inferior margin (surface) of the liver
Right	Cystic duct
Left	Common hepatic duct



CALOT’s triangle with the cystic artery, caudal view. Fig 6.115

CLINICAL REMARKS (reading)

- Repeated gallstone cholecystitis → **cholecystectomy**
- CALOT’s triangle = key landmark → identify all structures **before** ligating the **cystic artery** and **cystic duct** → avoids accidentally tying off the **common bile duct** (→ cholestasis)
- Contrast X-ray of the gallbladder and ducts (Figs 6.116–6.117) → shows ducts and **non-calcified stones**; **cholangiocarcinoma** or **pancreatic carcinoma** → cholestasis seen as **dilated ducts**

MEMORY AID (not from reading)

- CALOT borders = “**3 C’s**” → **C**ystic duct, **C**ommon hepatic duct, **C**ystic artery’s home under the liver (some books use the cystic artery itself as the upper border)

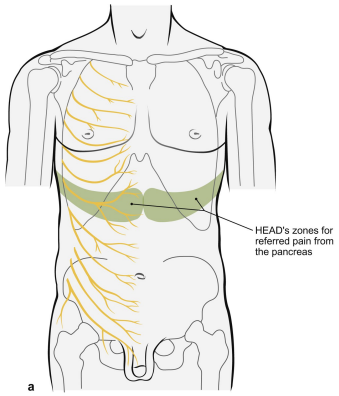
RECAP — LIVER AND BILIARY TREE

- Liver: 1,200–1,800 g; top at right 4th ICS / left 5th rib; lower edge at the costal margin; referred pain T8–T11 + C4
- Bare areas: bare area, porta, gallbladder bed, IVC groove · teres = umbilical vein, venosum = ductus venosus
- Lobule (central vein) · portal lobule (bile duct) · acinus (zones) · 8 segments; functional border = IVC–gallbladder plane
- Portal vein 75 % / hepatic artery 25 %; SMV + splenic vein behind pancreatic neck; IMV → splenic vein 70 %
- Portocaval: oesophageal (deadly), paraumbilical (caput medusae), rectal, retroperitoneal
- Gallbladder 40–70 ml; CBD 6 cm, > 1 cm = cholestasis; ampulla in 60 %; papilla 8–10 cm from pylorus
- CALOT: liver above, cystic duct right, common hepatic duct left; cystic artery inside in 75 %

C. PANCREAS

3.14 Projection and Referred Pain of the Pancreas [§6.6.1 · Figs 6.118–6.120]

- **Secondarily retroperitoneal** → projects onto **L1–L2**
- **Head** → against the descending duodenum · **body** → crosses the spine · **tail** → reaches the **splenic hilum**
- HEAD's zone → not sharply localised; pain usually referred to **T8**
- Special feature → because the pancreas is **retroperitoneal**, pain is also projected to the **back** in the same dermatome



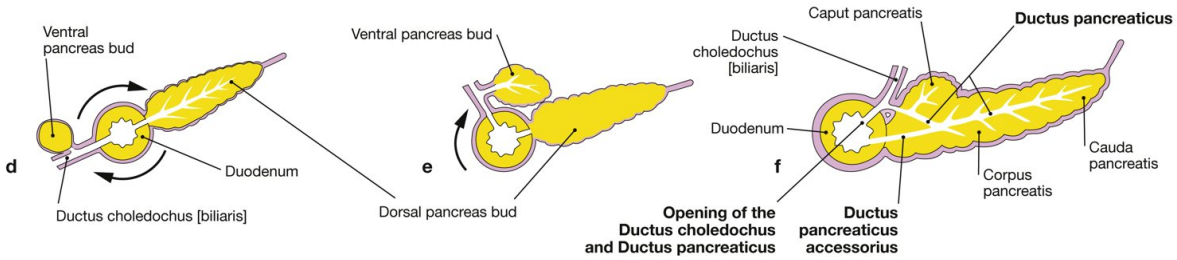
HEAD's zone of the pancreas (T8), ventral view; the dorsal view in the reading shows the same band on the back. Fig 6.120

CLINICAL REMARKS (reading)

- **Pancreatitis** → most often from a **gallstone blocking the papilla** (backflow of pancreatic juice) or **alcohol** → typical **belt-like** radiating pain
- **Pancreatic carcinoma** → pain radiating to the **back** → tumour infiltrates the aortic autonomic plexus (**Plexus aorticus abdominalis**)

3.15 Development and Malformations [§6.6.2 · Figs 6.121–6.122]

- **Day 28** → **ventral** and **dorsal pancreatic buds** from foregut endoderm, below the liver/gallbladder primordium, at the duodenum
- Ventral bud → **rotates dorsally** (with the duodenum) → **fuses with the dorsal bud in weeks 6–7**
- Main pancreatic duct → **distal dorsal duct + ventral duct** → opens at the **major papilla**
- Proximal part of the dorsal duct → may persist as the **accessory pancreatic duct** → **minor papilla**



Pancreatic development: ventral bud rotates behind the duodenum and fuses with the dorsal bud. Fig 6.121

Malformation	What happens	Clinical effect (reading)
Annular pancreas (Pancreas anulare)	Ring of pancreatic tissue around the **descending duodenum**	Food passage blocked → **ileus with vomiting**, especially newborns; often only when switching from milk to **solid food** → duodenum cut and re-sewn beside the duct system

Malformation	What happens	Clinical effect (reading)
Pancreas divisum	Ducts **fail to fuse** → main and accessory ducts open separately	Dorsal duct may be the main drainage (**10 %**) → secretion stasis → **recurrent pancreatitis**; consider after excluding gallstones and alcohol
Stenosed accessory duct	Normal fusion, but accessory duct narrowed at its opening (Fig 6.122b)	—

CLINICAL REMARKS (reading)

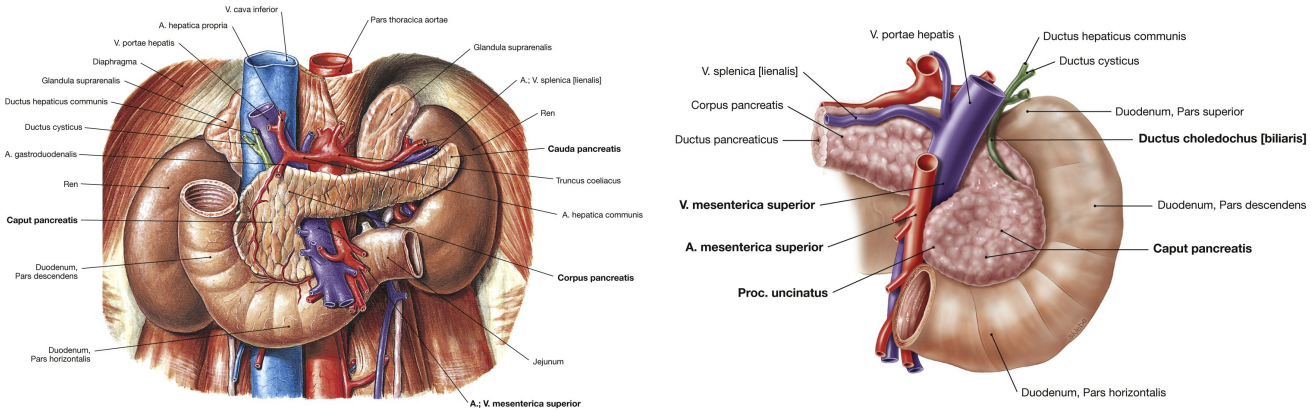
- Normally (ducts fused) → if a gallstone blocks the major papilla, pancreatic juice can still escape through the **minor papilla**
- In **pancreas divisum** this escape route is lost for the ventral part → **cholestasis plus severe pancreatitis**

HEADS-UP — SOURCE NOTE

- The reading gives two different figures for the accessory pancreatic duct: **6 %** in the development text (§6.6.2) but **65 %** in the duct section (Fig 6.126/6.127f). For exams, the duct-anatomy figure (**65 %**) is the one usually quoted; the 6 % may be a typo.

3.16 Structure, Relations and Histology [§6.6.3–6.6.4 · Figs 6.123–6.125]

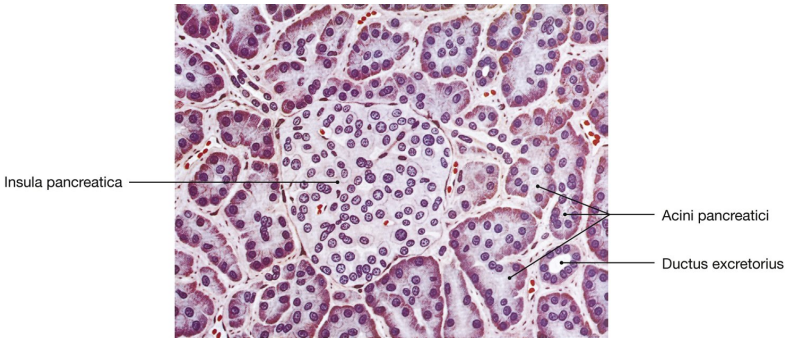
- **Head** → in the C-loop of the duodenum; **uncinate process** hooks **behind the SMA and SMV**; horizontal duodenum lies below
- **Neck** → short (**1.5–2 cm**), **in front of the SMA/SMV**
- **Body** → crosses the spine · **Tail** → behind the left colic flexure, in front of the **left kidney**, to the **splenic hilum**
- Surfaces → anterior + posterior, separated by superior and inferior margins
- **Anterior surface** → covered by parietal peritoneum → forms the **posterior wall of the omental bursa**
- **Posterior surface** → fused to the original parietal peritoneum → seen at dissection as **TOLDT's fascia**
- Common bile duct → **obliquely pierces** the head on its way to the major papilla (dorsal view)



Left: retroperitoneal organs of the epigastrium — pancreas, duodenum, kidneys, adrenals (Fig 6.123). Right: pancreas and duodenum, dorsal view; common bile duct in the head, uncinate process around the SMA/SMV (Fig 6.124)

Exocrine and endocrine parts

- **Mixed gland**
- **Exocrine** → **acini** make digestive enzymes as **inactive precursors** → duct system → intestine
- **Endocrine** → **islets of LANGERHANS**, scattered in the exocrine tissue, **most numerous in the tail** → **insulin**, **glucagon** and other hormones → blood → regulate blood glucose



Pancreas, microscopic view: exocrine acini around an islet of Langerhans. Fig 6.125

CLINICAL REMARKS (reading)

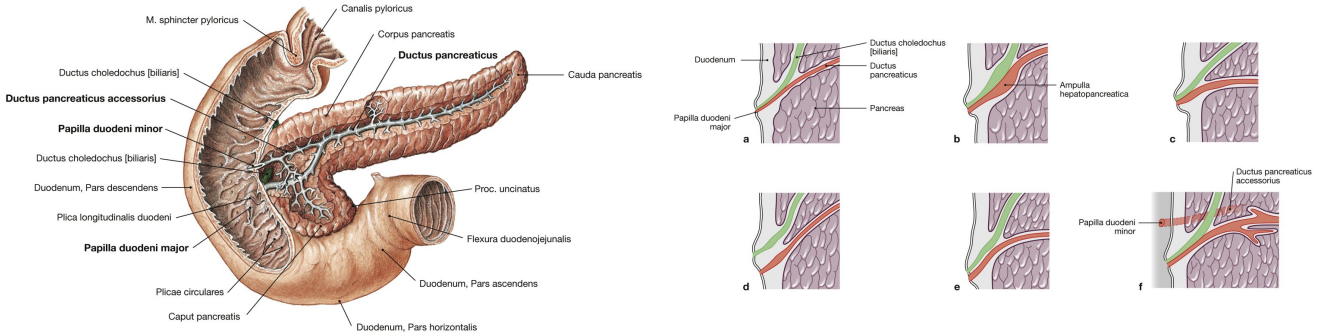
- ERCP near the head → close to the **SMA/SMV and portal vein** → vessel injury, usually needing **emergency surgery**
- Pancreatic necrosis (e.g. pancreatitis) → **maldigestion, diarrhoea**; loss of **80–90 %** of tissue → **diabetes mellitus**

CLINICAL CORRELATION (not from reading)

- Tail surgery (e.g. distal pancreatectomy) removes the islet-rich part → higher risk of diabetes

3.17 Excretory Ducts of the Pancreas [§6.6.5 · Figs 6.126–6.127]

- **Main duct (WIRSUNG)** → joins the common bile duct in **60 %** → **hepatopancreatic ampulla** → **major papilla (VATER)**
- **Accessory duct (SANTORINI)** → present in **65 %** → opens separately about **2 cm proximal** to the major papilla (minor papilla)
- Six junction variants (Fig 6.127): long common portion · ampullary dilation (**60 %**) · short common portion · separate openings · single opening with a septum · additional accessory duct (**65 %**)



Left: pancreatic duct system (Fig 6.126). Right: variants of the junction of the pancreatic and common bile ducts (Fig 6.127)

CLINICAL REMARKS (reading)

- Commonest causes of pancreatitis → **alcohol** and **gallstone damage at the papilla** → backflow of secretions → **autodigestion**
- An accessory duct with its own opening that **communicates with the main duct** → can act as a **safety outlet**

MEMORY AID (not from reading)

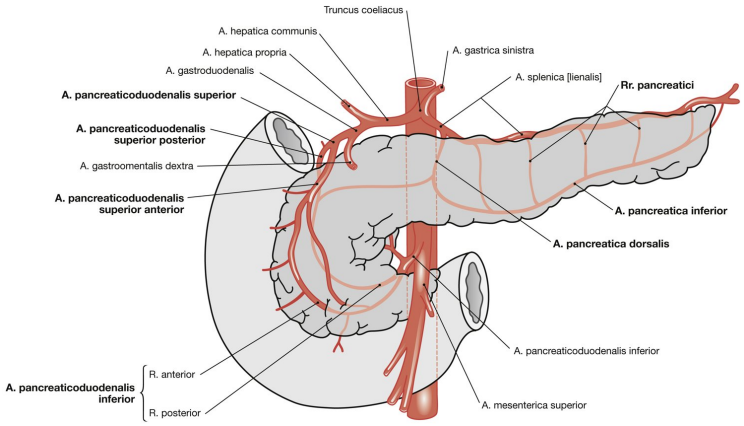
- **W**irsung = **W**ide main duct → major papilla · **S**antorini = **S**mall accessory duct → minor papilla, **S**uperior (proximal)

3.18 Arteries, Lymph and Nerves of the Pancreas [§6.6.6–6.6.8 · Figs 6.128–6.131]

Arteries

Region	Arteries	Origin
Head and neck	**Double arcades** → superior pancreaticoduodenal aa. (anterior + posterior) and inferior pancreaticoduodenal a. (anterior + posterior branch)	Superior → **gastroduodenal a.** (coeliac trunk) · inferior → **SMA**
Body and tail	**Pancreatic branches** → form the dorsal pancreatic a. and inferior pancreatic a.	**Splenic artery** (coeliac trunk)

- Inferior pancreatic a. → usually linked to the posterior arcades of the head → **marked redundancy**
- Veins → follow the arteries → **SMV** and **splenic vein** → portal vein



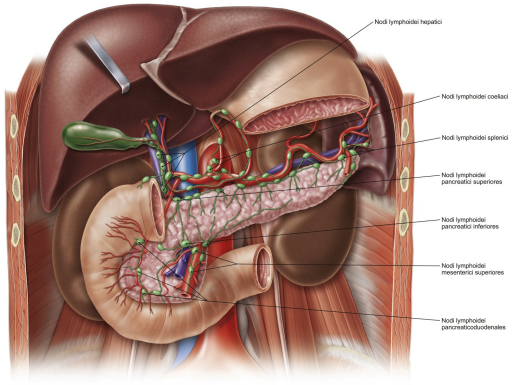
Arteries of the pancreas: pancreaticoduodenal arcades for the head, splenic branches for body and tail. Fig 6.128

CLINICAL REMARKS (reading)

- Two coeliac sources + the SMA → **pancreatic infarction is rare**

Lymph drainage

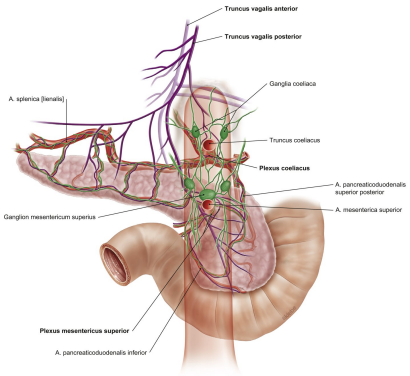
Region	Regional nodes	Onward drainage
Head and neck	Anterior + posterior **pancreaticoduodenal** nodes	Hepatic nodes → coeliac nodes, or directly → superior mesenteric nodes → intestinal trunks → thoracic duct
Body	Superior + inferior **pancreatic** nodes (along splenic a./v.)	Upper edge → coeliac nodes · lower edge → superior mesenteric nodes · also **lumbar** nodes (retroperitoneal) → lumbar trunks
Tail	**Splenic** nodes	Along splenic vessels → coeliac nodes



Lymph vessels and nodes of the pancreas. Fig 6.129

CLINICAL REMARKS (reading)
- Many interconnected routes → pancreatic carcinoma usually has **extensive nodal metastases at diagnosis** → curative surgery seldom possible

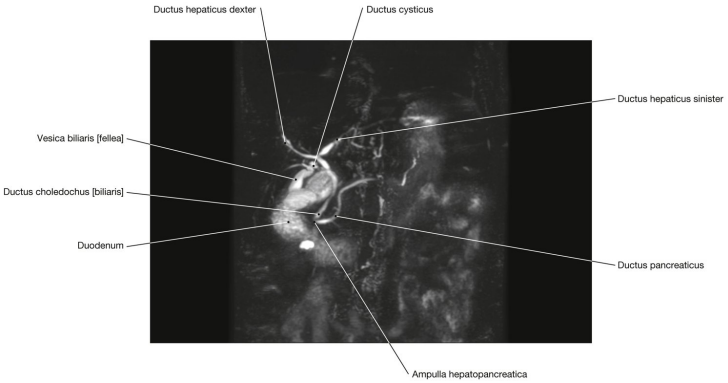
Innervation
- **Parasympathetic** → **promotes** enzyme secretion and insulin release · **Sympathetic** → **inhibits** them
- Sympathetic postganglionic + parasympathetic preganglionic fibres → from the **coeliac plexus** via **perivascular plexuses**
- Head → also direct fibres from the **posterior vagal trunk** (sometimes the anterior)
- Parasympathetic synapses → in **tiny ganglia within the pancreas**



Autonomic innervation of the pancreas, dorsal view. Fig 6.131

CLINICAL REMARKS (reading)
- Pancreatic cancer often becomes symptomatic **only after invading the aortic plexus** → **intense back pain** → complete removal impossible → **poor prognosis**

3.19 Pancreas Imaging [§6.6.9 · Figs 6.132–6.133]
- **Ultrasound** first → e.g. swelling in pancreatitis; often **unsatisfactory** because the retroperitoneal pancreas is hidden behind **air-filled intestine**
- Unclear → **CT**
- **ERCP** → contrast through the major papilla fills the pancreatic duct and common bile duct
  - Diagnoses **pancreas divisum** as a cause of recurrent pancreatitis
  - **Interrupted ducts** → suggest a **malignant tumour**



ERCP: pancreatic duct and common bile duct filled with contrast. Fig 6.133

HEADS-UP — SOURCE NOTE
- The imaging remark blames “air-filled **lungs**” for poor ultrasound views, but the figure legend (Fig 6.132) correctly says air-filled **intestines**. Read it as intestines.

RECAP — PANCREAS

- Secondarily retroperitoneal, L1–L2; referred pain T8, front and back; pancreatitis → belt-like pain
- Ventral bud rotates and fuses with dorsal bud (weeks 6–7); annular pancreas → duodenal obstruction; divisum → recurrent pancreatitis
- Uncinate process behind SMA/SMV; neck in front of them; anterior surface = back wall of omental bursa; TOLDT's fascia behind
- Wirsung + CBD → ampulla (60 %) → major papilla; Santorini (65 %) → minor papilla 2 cm proximal
- Head: pancreaticoduodenal arcades (GDA + SMA); body/tail: splenic artery; infarcts rare
- Islets mostly in the tail; 80–90 % loss → diabetes

D. SPLEEN

3.20 Projection, Referred Pain and Peritoneal Ligaments [§6.6.10–6.6.11 · Figs 6.134–6.137]

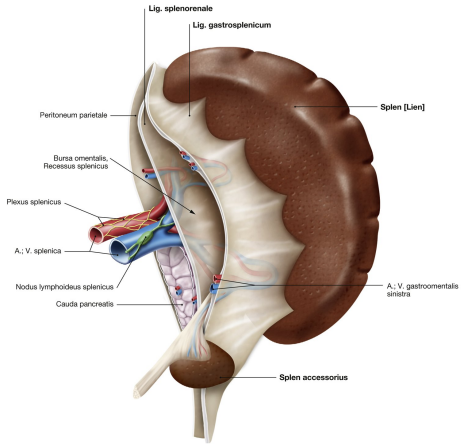
- **Intraperitoneal**, left upper abdomen; long axis along the **10th rib**
- A normal-sized spleen is **not palpable** below the costal arch
- Large contact with the diaphragm → position **moves with breathing**
- Lies in the **splenic niche** → floored by the **phrenicocolic ligament** (left colic flexure ↔ diaphragm)
- HEAD's zone → **T8–T9**, left upper abdomen → pain with **swelling or rupture**

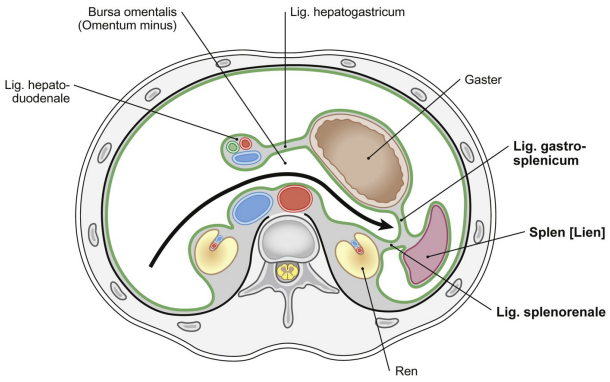


Left: projection of the spleen, left lateral view (Fig 6.134c). Right: HEAD's zone of the spleen, T8–T9 (Fig 6.135)

Peritoneal ligaments

- Two peritoneal folds, both inserting at the **splenic hilum**:
  - **Gastrosplenic ligament** (ventral) → from the stomach
  - **Splenorenal ligament** (dorsal) → to the posterior trunk wall
- Between them → **splenic recess** of the **omental bursa**, reaching the hilum
- Behind the splenorenal ligament (retroperitoneal) → **splenic artery** (+ splenic plexus), **splenic vein**, lymph vessels → enter **above the pancreatic tail**





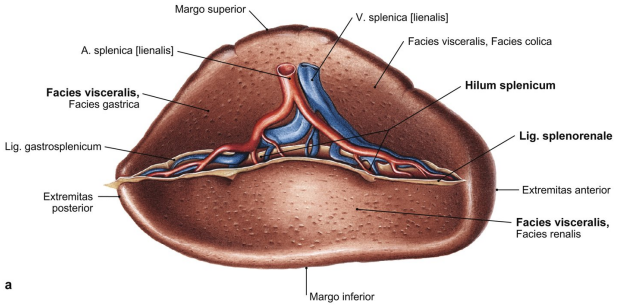
Left: spleen folded back to show the splenorenal ligament and hilar vessels (Fig 6.136). Right: cross-section — omental bursa (arrow) reaching the hilum between the gastrosplenic and splenorenal ligaments (Fig 6.137)

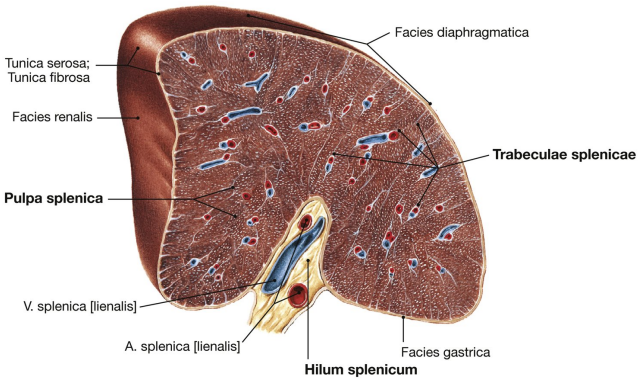
MEMORY AID (not from reading)

- Spleen "**1, 3, 5, 7, 9, 11**" (classic, not from reading) → 1 × 3 × 5 inches, 7 oz, ribs 9–11 → fits the reading's 150 g, 11 × 7 × 4 cm and 10th rib
- Ligament order front → back: "**Gastric Gate, Renal Rear**" → gastrosplenic in front, splenorenal behind

3.21 Structure and Circulation of the Spleen [§6.6.12 · Figs 6.138–6.140]

- **Secondary lymphoid organ** → immune function + **blood filtering**
- Size → **150 g**, **11 cm long, 7 cm wide, 4 cm thick**
- **Diaphragmatic surface** → convex · **visceral surface** → concave, touches **left kidney, left colic flexure, stomach**
- **Superior margin → notched** · inferior margin → smooth
- Vessels branch at the hilum → **segments**, not visible on the surface
- Firm **capsule** → **trabeculae** into the pulp, carrying larger artery and vein branches



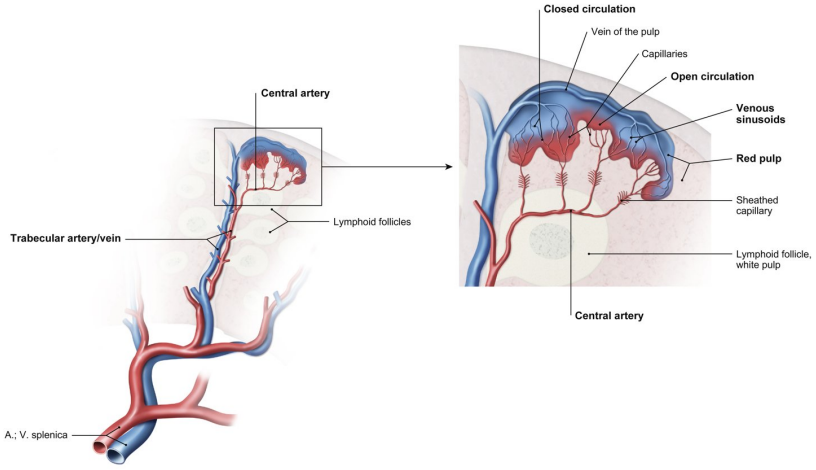


Left: spleen, medial ventral view — hilum, visceral surfaces, notched superior margin (Fig 6.138a). Right: section through the hilum — capsule, trabeculae, splenic pulp (Fig 6.139)

	Red pulp	White pulp
Appearance	Blood-filled fibrous meshwork	Scattered white nodules (microscopic)
Contents	Blood from the open circulation; sinusoids	Lymphatic tissue → **lymphatic follicles** + **periarterial lymphatic sheaths (PALS)**
Functions	Breaks down old/abnormal **red cells**; **stores platelets**; with the liver → **fetal blood formation**	Immune responses (secondary lymphoid tissue)

Vessel path through the spleen

- Splenic artery → branches at the hilum → **trabecular arteries**
- Terminal branches **do not anastomose** → **functional end arteries** → define the segments
- Trabecular arteries → **central arteries** (sheathed by white pulp lymphocytes) → brush-like branches → capillaries
- Capillaries end in two ways:
  - **Open circulation** → blood pours into the red pulp mesh → cells must squeeze back **between endothelial cells of the sinusoids** → **old or abnormal red cells are trapped and removed**
  - **Closed circulation** → directly into venous sinusoids
- Sinusoids → pulp veins → trabecular veins → **splenic vein**



Parenchyma and vessels of the spleen: trabecular artery → central artery in white pulp (PALS, follicle) → open and closed circulation → sinusoids → trabecular vein. Fig 6.140

CLINICAL REMARKS (reading)

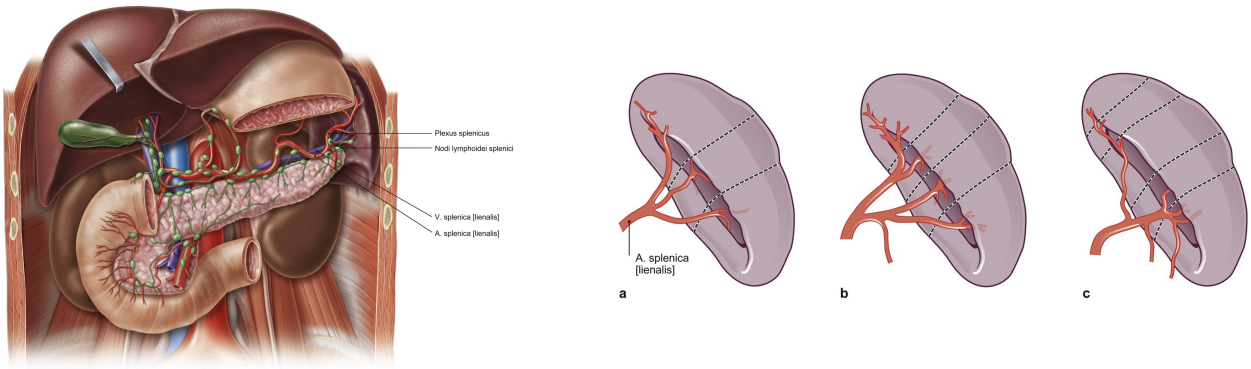
- High under the ribs → covered by ribs even in inspiration
- **Splenomegaly** (several kilograms) → **leukaemia, lymphoma**, viral infections such as **infectious mononucleosis** ("kissing disease", spread by saliva) → palpable mass in the left upper abdomen

MEMORY AID (not from reading)

- Open circulation = "**squeeze test**": red cells that cannot squeeze back into the sinusoids fail and are eaten

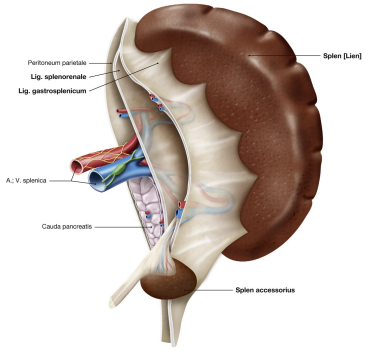
3.22 Splenic Vessels, Segments and Accessory Spleen [§6.6.13–6.6.14 · Figs 6.141–6.143]

- **Splenic artery** → from the **coeliac trunk** → turns left, runs retroperitoneally along the **upper border of the pancreas** (with the splenic plexus)
- At the hilum → **2–3 main** branches → up to **6 terminal** branches
- Branches to the stomach:
  - **Posterior gastric artery** → in **30–60 %**, before branching
  - At the hilum → **left gastro-omental artery** + **short gastric arteries**
- **Splenic vein** → behind the pancreas to the neck → joins the **SMV** → portal vein; receives the **IMV in 70 %**
- **Splenic nodes** at the hilum → drain the spleen, **pancreatic tail** and upper **greater curvature** → coeliac nodes → intestinal trunks



Left: neurovascular pathways of the spleen, ventral view (Fig 6.141). Right: three, four or five splenic segments supplied by end arteries (Fig 6.142)

- **Segments** → terminal arteries are end arteries → **3–6 wedge-shaped** segments
- **Accessory spleen** → **5–30 %** (**< 15 %** in large real-world studies) → usually near the hilum, in a peritoneal fold (e.g. splenorenal ligament)



Accessory spleen in the splenorenal ligament. Fig 6.143

CLINICAL REMARKS (reading)

- **Splenic rupture** (abdominal trauma) → life-threatening bleeding
  - **Vertical tears** crossing several segments → bleed **heavily**
  - **Horizontal tears** (along segment borders) → bleed **relatively little**
- End arteries → splenic **infarcts are wedge-shaped**, between segment borders
- **Splenectomy** → splenic artery and vein removed; spleen stores many platelets → **prevent blood clots** around surgery (stroke, heart attack risk)
- Accessory spleen after splenectomy for **trauma** → can take over → immune function kept
- Splenectomy for **haemolytic anaemia** (abnormal red cells destroyed too fast) → **remove the accessory spleen too**, or symptoms persist

CLINICAL CORRELATION (not from reading)

- After splenectomy → platelet count rises (reactive thrombocytosis) and there is lifelong risk of overwhelming infection by encapsulated bacteria → vaccinate against pneumococcus, meningococcus and Hib

RECAP — SPLEEN

- Intraperitoneal, along the 10th rib, not palpable; splenic niche over the phrenicocolic ligament; referred pain T8–T9
- Gastrosplenic (front) and splenorenal (back) ligaments; splenic recess of omental bursa between them
- 150 g, 11 × 7 × 4 cm; notched superior margin; red pulp filters RBCs and stores platelets; white pulp = follicles + PALS
- Open vs closed circulation; end arteries → 3–6 segments → wedge infarcts; vertical tears bleed more
- Accessory spleen 5–30 % → remove it in splenectomy for haemolysis, keep it after trauma

PART 4 Comparison Tables

Table 1 — The four organs at a glance

	Liver	Gallbladder	Pancreas	Spleen
Peritoneum	Intraperitoneal (bare areas)	Intraperitoneal	**Secondarily retroperitoneal**	Intraperitoneal
Projection	Top: R 4th ICS / L 5th rib; edge at costal margin	Fundus: R MCL, 9th rib	L1–L2	Along 10th rib (left)
Size	1,200–1,800 g; ≤ 12 cm in R MCL	40–70 ml	Neck 1.5–2 cm	150 g; 11 × 7 × 4 cm
HEAD's zone	T8–T11 right + C4 (shoulder)	T8–T11 right + C4	T8 (front **and back**)	T8–T9 left
Palpable normally?	No	No	No	No

Table 2 — Embryology

Structure	Origin	Becomes
Hepatic diverticulum (day 22, week 4)	Foregut endoderm at future duodenum	Upper part → liver · lower part → bile ducts + gallbladder
Septum transversum	Mesoderm	Liver connective tissue, sinusoids, haematopoietic islands
Ventral mesohepaticum	Ventral mesogastrium	Falciform ligament
Dorsal mesohepaticum	Ventral mesogastrium	Lesser omentum
Umbilical vein · ductus venosus	Fetal circulation	Round ligament · ligamentum venosum
Ventral + dorsal pancreatic buds (day 28)	Foregut endoderm	Fuse in weeks 6–7 → pancreas; main duct = distal dorsal + ventral duct

Table 3 — Anatomical vs functional liver lobes

	Anatomical lobes	Functional lobes
Dividing line	Falciform ligament (front); round lig. / lig. venosum fissure (back)	IVC–gallbladder plane (no surface marker)
Left contains	Segments II–III only	Segments II–IV
Right contains	Segments IV–VIII (+ quadrate and caudate lobes)	Segments V–VIII
Basis	Surface shape	Branches of the portal triad
Segment I (caudate)	Part of the anatomical right lobe	Supplied from both sides → autonomous

Table 4 — Blood supply in and out of the liver

Vessel	Share	Character	Route
Portal vein	75 %	Nutrient-rich	SMV + splenic vein behind pancreatic neck → hepatoduodenal ligament → porta
Proper hepatic artery	25 %	Oxygen-rich	Coeliac trunk → common hepatic a. → (right gastric a.) → proper hepatic a. → right + left branches
Hepatic veins (3)	Outflow	Mixed, deoxygenated	Central veins → sublobular veins → hepatic veins → IVC
Short hepatic veins	Outflow	—	Caudate lobe → directly to IVC

Table 5 — Lymph drainage of the three glands

Organ	Main route	Other routes	Clinical point
Liver	Caudal → hepatic → coeliac nodes → intestinal trunk (80 %)	Cranial → phrenic → mediastinal nodes · round ligament → inguinal/axillary · left lobe → stomach/pancreas	Thoracic node spread; Sister Mary Joseph node
Gallbladder	Cystic node → hilar nodes	—	—
Pancreas	Head: pancreaticoduodenal · body: pancreatic · tail: splenic → coeliac / sup. mesenteric	Lumbar nodes (retroperitoneal)	Extensive nodal spread at diagnosis
Spleen	Splenic nodes → coeliac → intestinal trunks	Also drain pancreatic tail + upper greater curvature	—

Table 6 — Duct anatomy: bile and pancreatic ducts

Duct	Formed by / course	Numbers	Opens at
Cystic duct	Gallbladder neck; spiral fold (Heister)	~3 mm × 2–4 cm	Joins common hepatic duct
Common hepatic duct	Right + left hepatic ducts	—	Joins cystic duct
Common bile duct	Hepatoduodenal lig. (front of portal v.) → behind sup. duodenum → pancreatic head	~6 cm × 0.4–0.9 cm; > 1 cm = cholestasis	Major papilla (ampulla in 60 %)
Main pancreatic duct (Wirsung)	Distal dorsal + ventral ducts	Joins CBD in 60 %	Major papilla, 8–10 cm from pylorus
Accessory duct (Santorini)	Proximal dorsal duct	Present in 65 %	Minor papilla, 2 cm proximal

Table 7 — Portocaval anastomoses

Site	Portal → caval connection	Sign	Importance
Oesophagus	L + R gastric vv. → oesophageal vv. → azygos → SVC	Oesophageal varices	Life-threatening bleeding
Umbilicus	Paraumbilical vv. → epigastric / thoracoepigastric vv. → SVC + IVC	Caput medusae	Diagnostic
Rectum	Superior rectal v. → middle + inferior rectal vv. → internal iliac → IVC	Rectal veins dilated	Not significant (reading)
Retroperitoneum	IMV → testicular/ovarian vv. → IVC	—	Not significant

Table 8 — Arterial supply of the region

Target	Artery	Parent
Liver	Proper hepatic a. (right + left branches)	Common hepatic a. ← coeliac trunk (variants: SMA 10–20 % right, left gastric 10–20 % left)
Gallbladder	Cystic a. (in CALOT's triangle in 75 %)	Right branch of proper hepatic a.
CBD (upper)	Cystic a., right hepatic branch, gastroduodenal a.	—
CBD (distal third) + ampulla	Posterior superior pancreaticoduodenal a.	Gastroduodenal a.
Pancreatic head + neck	Sup. pancreaticoduodenal aa. + inf. pancreaticoduodenal a.	Gastroduodenal a. + SMA
Pancreatic body + tail	Pancreatic branches (dorsal + inferior pancreatic aa.)	Splenic a.
Spleen	Splenic a. → 2–3 main → up to 6 terminal branches	Coeliac trunk
Stomach (from splenic a.)	Posterior gastric (30–60 %), left gastro-omental, short gastric aa.	Splenic a.

Table 9 — Red vs white pulp; open vs closed circulation

	Red pulp / open circulation	White pulp / closed circulation
Structure	Fibrous mesh filled with blood	Lymphatic follicles + PALS around central arteries
Blood route	Capillaries open into mesh → cells re-enter sinusoids between endothelial cells	Capillaries pass straight into venous sinusoids
Function	Remove old/abnormal RBCs, store platelets, fetal haematopoiesis	Immune response

Table 10 — Referred pain (HEAD's zones)

Organ	Dermatome(s)	Why / extra
Liver + gallbladder	Right T8–T11 + C4 (right shoulder)	Phrenic nerve (C4) → R. phrenicoabdominalis to liver/gallbladder peritoneum
Pancreas	T8, front **and back**	Retroperitoneal position → back projection
Spleen	Left T8–T9	Swelling or rupture

PART 5 High-Yield Exam Points

1. Liver and gallbladder → intraperitoneal, right upper abdomen; upper margin right 4th ICS, left 5th rib
2. Gallbladder fundus → right midclavicular line at the 9th rib
3. Normal liver edge follows the costal margin → not palpable; ≤ 12 cm in the right MCL
4. Liver/gallbladder pain → right T8–T11 + right shoulder (C4, phrenic nerve)
5. Liver + bile ducts → foregut endoderm, hepatic diverticulum from day 22; grows into the septum transversum
6. Ventral mesohepaticum → falciform ligament; dorsal mesohepaticum → lesser omentum
7. Liver = largest gland, 1,200–1,800 g
8. Round ligament = umbilical vein; ligamentum venosum = ductus venosus
9. Four peritoneum-free areas → bare area, porta hepatis, gallbladder bed, IVC groove
10. Porta hepatis → portal vein, proper hepatic artery, common hepatic duct
11. Quadrate lobe ventral, caudate lobe dorsal on the visceral surface
12. GLISSON's triad → interlobular artery, vein and bile duct in connective tissue
13. Classic lobule → central vein centre; portal lobule → bile duct centre; acinus → zones by oxygen supply
14. 3 hepatic veins → 4 divisions; portal triad branches → 8 independent segments
15. Functional left lobe II–IV, right lobe V–VIII; border = IVC–gallbladder plane, not falciform
16. Segment I (caudate) → supplied from both sides; drains directly into IVC
17. Segmental resection → little blood loss; segment identified by ligating its vessel
18. Portal vein 75 % (nutrients); hepatic artery 25 % (oxygen)
19. Coeliac trunk → common hepatic → (right gastric) → proper hepatic → right/left; cystic a. from right branch
20. Textbook arterial pattern ~50 %; SMA to right lobe or left gastric to left lobe 10–20 % each
21. Liver lymph → 80 % caudal to hilum → coeliac; cranial route → mediastinal nodes → thoracic metastases
22. Sister Mary Joseph node → umbilical metastasis via round ligament lymphatics
23. Portal vein = SMV + splenic vein behind pancreatic neck; ~8 cm; IMV → splenic vein 70 %
24. Portocaval: oesophageal, paraumbilical, rectal, retroperitoneal
25. Ruptured oesophageal varices → most common cause of death in cirrhosis; caput medusae → diagnostic
26. Cirrhosis causes → worldwide hepatitis B/C/D; industrialised world alcohol, steatosis (diabetes, obesity)
27. Liver biopsy → ultrasound-guided, in expiration, low ICS, above the rib below; risk pneumothorax; shoulder pain
28. Gallbladder 40–70 ml; spiral fold of Heister in cystic duct
29. Cystic duct ~3 mm × 2–4 cm; CBD ~6 cm × 0.4–0.9 cm; CBD > 1 cm → cholestasis
30. Cholecystitis → sepsis risk → laparoscopic cholecystectomy within 24 h
31. Ampulla forms in 60 %; papilla 8–10 cm from pylorus, middle third of descending duodenum
32. Sphincter of Oddi → around the ampulla; separate sphincters on the CBD and pancreatic duct
33. Distal CBD + ampulla → posterior superior pancreaticoduodenal artery
34. Stone at papilla → painful swollen gallbladder + jaundice; painless big gallbladder → think pancreatic head cancer
35. CALOT's triangle → liver above, cystic duct right, common hepatic duct left; cystic artery inside in 75 %
36. Pancreas → secondarily retroperitoneal, L1–L2; pain T8 front and back; pancreatitis → belt-like pain
37. Pancreatitis → gallstones and alcohol; cancer → back pain from aortic plexus invasion
38. Ventral + dorsal buds (day 28) fuse in weeks 6–7
39. Annular pancreas → duodenal obstruction (newborns, at weaning); divisum → recurrent pancreatitis
40. Uncinate process behind SMA/SMV; neck in front of them; tail reaches splenic hilum
41. Anterior pancreas = posterior wall of omental bursa; posterior fusion = TOLDT's fascia
42. Islets of Langerhans mostly in the tail; 80–90 % loss → diabetes

43. Wirsung → major papilla; Santorini (65 %) → minor papilla 2 cm proximal
44. Pancreatic head → gastroduodenal + SMA arcades; body/tail → splenic artery → infarction rare
45. Pancreatic cancer → extensive nodal spread at diagnosis; curative surgery rare
46. Parasympathetic ↑ and sympathetic ↓ pancreatic secretion and insulin
47. Spleen → intraperitoneal, along 10th rib, not palpable; splenic niche over phrenicocolic ligament; pain T8–T9
48. Gastrosplenic (front) + splenorenal (back) ligaments; vessels behind the splenorenal ligament
49. Spleen 150 g, 11 × 7 × 4 cm; notched superior margin
50. Red pulp → RBC breakdown, platelet storage, fetal haematopoiesis; white pulp → follicles + PALS
51. Open circulation → RBCs squeeze back into sinusoids → old cells filtered out
52. Splenic end arteries → 3–6 segments → wedge infarcts; vertical tears bleed most
53. Splenic artery → posterior gastric (30–60 %), left gastro-omental, short gastric aa.
54. Accessory spleen 5–30 % → keep after trauma splenectomy, remove in haemolytic anaemia
55. Splenomegaly → leukaemia, lymphoma, infectious mononucleosis

PART 6 Common Confusions

Confusion	Clarification
Round ligament vs ligamentum venosum	Round (teres) = obliterated **umbilical vein**; venosum = obliterated **ductus venosus**
Anatomical vs functional left lobe	Anatomical left = II–III (left of falciform); functional left = II–IV (left of IVC–gallbladder plane)
Is the quadrate lobe part of the left liver?	Anatomically it sits on the right lobe; functionally it is segment IV → **left** liver
Caudate lobe: left, right or neither?	Reading contradicts itself; it gets **both** sides' supply and drains directly to the IVC → treat as autonomous segment I
Hepatic veins vs portal triad	Hepatic veins run **between** divisions and drain **out**; triad branches run **inside** segments and bring blood **in** (and bile out)
Classic lobule vs portal lobule centre	Classic → central vein; portal → bile duct
Common hepatic duct vs common bile duct	Hepatic = right + left hepatic ducts; bile duct (choledochus) = hepatic + cystic duct
Common hepatic artery vs proper hepatic artery	Common → from coeliac trunk; proper → after the right gastric a. branches off
Which border of CALOT's triangle is the cystic duct?	Cystic duct = **right**; common hepatic duct = **left**; liver = **cranial**
Painful vs painless enlarged gallbladder	Painful → stone with cholestasis; painless → tumour (e.g. pancreatic head), no inflammation
Where does the IMV drain?	Splenic vein in 70 %, SMV in 30 % (not directly into the portal vein)
Which portocaval site matters most?	Oesophageal varices (bleeding); caput medusae is only diagnostic; rectal and retroperitoneal not significant
Wirsung vs Santorini	Wirsung = main → major papilla; Santorini = accessory → minor papilla, 2 cm proximal
Accessory duct frequency: 6 % or 65 %?	Reading gives both; 65 % is the duct-anatomy figure
Uncinate process vs neck relative to SMA/SMV	Uncinate → **behind** the vessels; neck → **in front**
Is the pancreas primarily retroperitoneal?	No → **secondarily** retroperitoneal (fused, TOLDT's fascia)

Confusion	Clarification
Where are most islets?	Pancreatic **tail**
Spleen: intra- or retroperitoneal?	Intraperitoneal; only its vessels run retroperitoneally behind the splenorenal ligament
Which splenic tear bleeds more?	Vertical (crosses segments) > horizontal (follows segment borders)
Keep or remove an accessory spleen?	Keep after trauma splenectomy; remove when splenectomy treats haemolytic anaemia
Steatosis on ultrasound	Reading says “low echo”; standard teaching (not from reading) → fatty liver is bright (hyperechoic)

PART 7 Review Q&A

The reading contains no questions. Below are practice questions built only from the reading.

PRACTICE QUESTION 1

A patient with acute cholecystitis complains of pain at the tip of the right shoulder. Which nerve root carries this referred pain?

A. T6
B. T10
C. C4
D. L1
E. C7

Answer: C — C4

- Phrenic nerve (mainly C4) → R. phrenicoabdominalis supplies the peritoneum over the liver and gallbladder
- The other liver/gallbladder HEAD's zones are right T8–T11

PRACTICE QUESTION 2

Which structure is the remnant of the fetal ductus venosus?

A. Ligamentum teres hepatis
B. Falciform ligament
C. Ligamentum venosum
D. Lesser omentum
E. Coronary ligament

Answer: C — Ligamentum venosum (ARANTII)

- Round ligament = umbilical vein; falciform ligament = ventral mesohepaticum; lesser omentum = dorsal mesohepaticum

PRACTICE QUESTION 3

A surgeon resects segments II, III and IV. Which plane marks the right edge of this functional left liver?

A. Falciform ligament
B. Plane of the right hepatic vein
C. Plane from the IVC to the gallbladder
D. Fissure for the ligamentum venosum
E. Plane of the round ligament

Answer: C — Plane from the IVC to the gallbladder

- Functional left (II–IV) and right (V–VIII) lobes are divided by the IVC–gallbladder plane, not the falciform ligament

PRACTICE QUESTION 4

A man with alcoholic cirrhosis vomits large amounts of blood. Which portocaval connection is responsible?

A. Paraumbilical veins to epigastric veins
B. Superior rectal vein to middle rectal veins
C. Left gastric vein to oesophageal veins and the azygos system
D. IMV to testicular vein
E. Cystic vein to hepatic veins

Answer: C — Left gastric vein to oesophageal veins and the azygos system

- Oesophageal varices → rupture → commonest cause of death in cirrhosis
- Paraumbilical → caput medusae (diagnostic only)

PRACTICE QUESTION 5

Why is a percutaneous liver biopsy done during expiration through a lower intercostal space?

A. To avoid the gallbladder
B. To avoid puncturing the pleural cavity
C. To reach the caudate lobe
D. To avoid the portal vein
E. To reduce shoulder pain

Answer: B — To avoid puncturing the pleural cavity

- The pleural cavity covers the liver from above → pneumothorax risk
- The needle passes above the lower rib to spare the intercostal vessels and nerve

PRACTICE QUESTION 6

During cholecystectomy, which structure forms the LEFT border of CALOT's triangle?

A. Cystic duct
B. Common hepatic duct
C. Inferior margin of the liver
D. Cystic artery
E. Common bile duct

Answer: B — Common hepatic duct

- Cranial → liver; right → cystic duct; left → common hepatic duct
- Cystic artery runs inside the triangle in 75 %

PRACTICE QUESTION 7

A jaundiced patient has a large, non-tender, palpable gallbladder. What should be excluded first?

A. Acute cholecystitis
B. Carcinoma of the pancreatic head
C. Hepatitis A
D. Splenic rupture
E. Annular pancreas

Answer: B — Carcinoma of the pancreatic head

- Stone at the papilla → painful swollen gallbladder; tumour → enlargement without inflammation, mostly painless

PRACTICE QUESTION 8

A newborn vomits after feeds; imaging shows pancreatic tissue encircling the descending duodenum. What is the diagnosis?

A. Pancreas divisum
B. Annular pancreas
C. Accessory spleen
D. Stenosis of the accessory duct
E. Ectopic pancreas

Answer: B — Annular pancreas

- Ring of pancreas around the descending duodenum → ileus with vomiting; may appear when switching to solid food
- Pancreas divisum → recurrent pancreatitis instead

PRACTICE QUESTION 9

Which arteries supply the head of the pancreas?

A. Splenic artery branches only
B. Superior pancreaticoduodenal aa. (gastroduodenal a.) and inferior pancreaticoduodenal a. (SMA)
C. Left gastric artery
D. Proper hepatic artery
E. Inferior mesenteric artery

Answer: B — Superior and inferior pancreaticoduodenal arteries

- Double arcades from coeliac trunk (via GDA) and SMA → infarction is rare
- Body and tail → pancreatic branches of the splenic artery

PRACTICE QUESTION 10

After a car accident, a patient's spleen has a vertical tear crossing several segments. Why does it bleed heavily?

A. The splenic vein has no valves
B. It crosses several segments supplied by separate end arteries
C. The red pulp stores platelets
D. The capsule is absent
E. The white pulp is highly vascular

Answer: B — It crosses several segments supplied by separate end arteries

- Terminal splenic arteries are functional end arteries → segments
- Horizontal tears follow segment borders → bleed relatively little

PRACTICE QUESTION 11

A patient undergoes splenectomy for hereditary haemolytic anaemia. An accessory spleen is found in the splenorenal ligament. What should be done?

A. Leave it to preserve immunity
B. Remove it too
C. Biopsy it only
D. Ligate its artery only
E. Nothing; accessory spleens are non-functional

Answer: B — Remove it too

- Otherwise it keeps destroying red cells → symptoms persist
- After trauma splenectomy, by contrast, it can be left to take over immune function

PRACTICE QUESTION 12

Where are the islets of Langerhans most concentrated?

A. Head
B. Uncinate process
C. Neck
D. Body
E. Tail

Answer: E — Tail

- Islets lie scattered in the exocrine tissue, especially in the tail

PART 8 One-Page Summary

The **liver** and **gallbladder** lie intraperitoneally in the right upper abdomen. The liver's upper margin projects to the right 4th intercostal space and left 5th rib, its lower edge follows the costal margin so a normal liver is not palpable, and the gallbladder fundus lies at the right midclavicular line on the 9th rib. Pain is referred to right T8–T11 and, via the phrenic nerve, to the C4 skin of the right shoulder. Both develop from foregut endoderm as the hepatic diverticulum (day 22) growing into the septum transversum; the ventral mesohepaticum becomes the falciform ligament and the dorsal one the lesser omentum. The liver (1,200–1,800 g) has diaphragmatic and visceral surfaces, anatomical right and left lobes divided by the falciform ligament, and quadrate and caudate lobes around the porta hepatis, where the portal vein and proper hepatic artery enter and the common hepatic duct leaves. The round ligament is the old umbilical vein and the ligamentum venosum the old ductus venosus; the bare area, porta, gallbladder bed and IVC groove lack peritoneum. Microscopically, hepatocyte plates radiate around a central vein (classic lobule), bile drains towards portal tracts (portal lobule), and oxygen supply defines zones within the acinus. Three hepatic veins divide the liver into four divisions and portal triad branches into eight independent segments: II–IV form the functional left liver and V–VIII the right, separated by the IVC–gallbladder plane, while the caudate lobe (I) has dual supply and drains directly to the IVC. The portal vein (75 % of flow), formed by the SMV and splenic vein behind the pancreatic neck, brings nutrient-rich blood, and the proper hepatic artery (25 %) brings oxygen; arterial variants from the SMA or left gastric artery occur in 10–20 % each. Most lymph drains via the hilum to coeliac nodes, but a cranial route reaches the mediastinum. In **cirrhosis**, scarring and pseudonodules raise resistance, causing portal hypertension that opens oesophageal (deadly varices), paraumbilical (caput medusae), rectal and retroperitoneal anastomoses. Biopsy is done under ultrasound in expiration through a low intercostal space above the rib to avoid the pleura and intercostal bundle. The **gallbladder** holds 40–70 ml; its cystic duct (spiral fold of Heister) joins the common hepatic duct to form the 6 cm common bile duct, which passes behind the duodenum and through the pancreatic head to join the pancreatic duct at the hepatopancreatic ampulla (60 %) and the major papilla, guarded by the sphincter of Oddi. A duct over 1 cm suggests cholestasis; a painless enlarged gallbladder suggests pancreatic head cancer. CALOT's triangle (liver, cystic duct, common hepatic duct) holds the cystic artery in 75 % and is the key landmark in cholecystectomy. The **pancreas** is secondarily retroperitoneal at L1–L2, with referred pain at T8 front and back. Its ventral bud rotates and fuses with the dorsal bud; failures give annular pancreas (duodenal obstruction) or pancreas divisum (recurrent pancreatitis). The uncinate process lies behind and the neck in front of the SMA/SMV, and islets cluster in the tail. Wirsung's duct opens at the major papilla and Santorini's (65 %) at the minor papilla. The head is supplied by pancreaticoduodenal arcades from the gastroduodenal artery and SMA, the body and tail by the splenic artery, so infarction is rare; wide lymph spread and invasion of the aortic plexus make cancer late and painful. The **spleen** is intraperitoneal along the 10th rib, held by gastrosplenic and splenorenal ligaments, weighs 150 g and refers pain to left T8–T9. Red pulp filters old red cells and stores platelets through an open circulation, white pulp forms follicles and PALS, and end arteries create 3–6 segments, so infarcts are wedge-shaped and vertical tears bleed most. An accessory spleen (5–30 %) is kept after trauma but removed when splenectomy treats haemolytic anaemia.